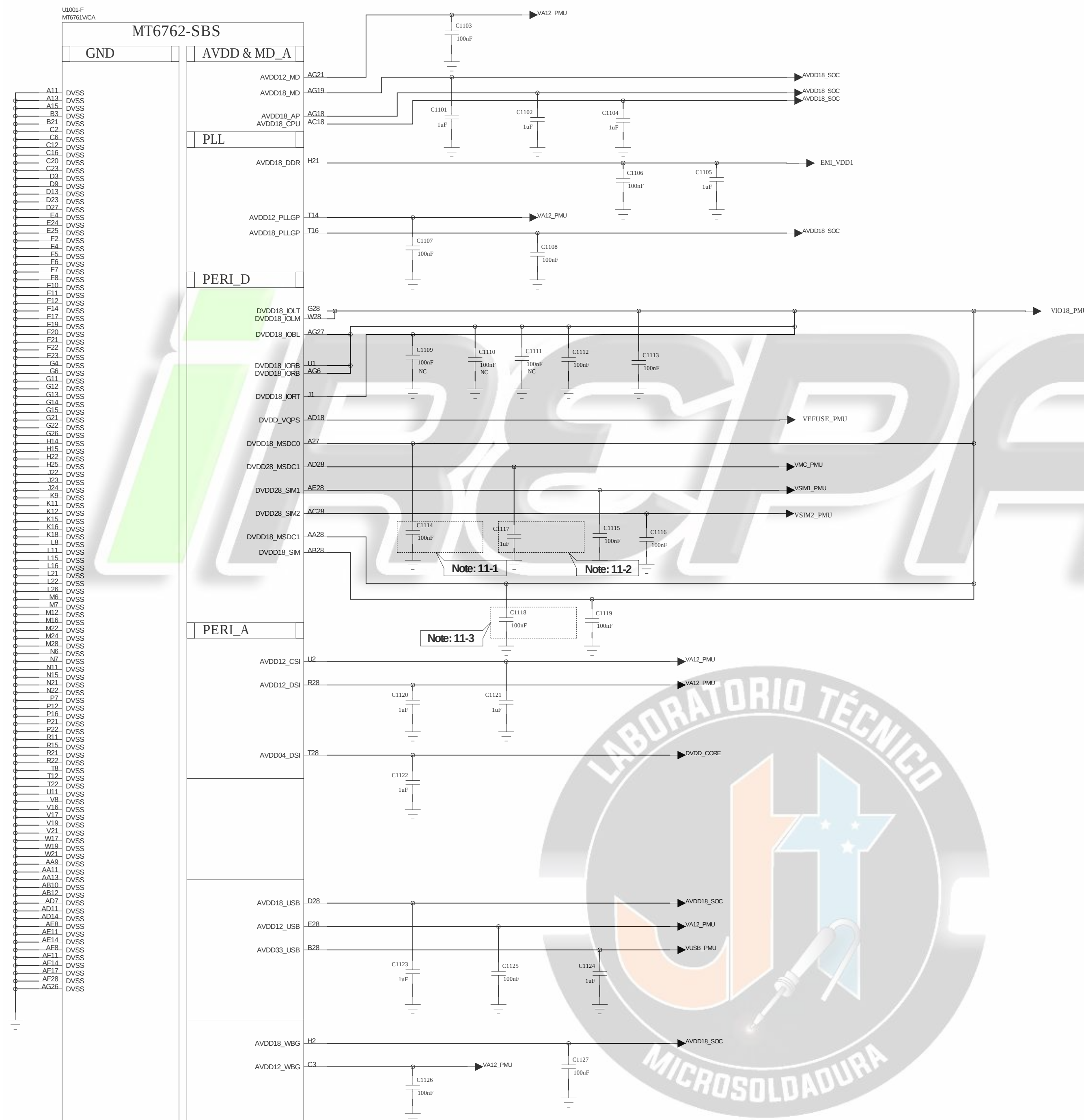


REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: C1114 closed DVDD18_MSDC0 150mil

Note 11-2: C1117 closed DVDD28_MSDC1 150mil

Note 11-3: C1118 closed DVDD18_MSDC1 150mil

DRAWN: Chunshen.zhou CHECKED: <Checked By> QUALITY CONTROL: <QC By> RELEASED: <Released By>		DATED: 20170417 DATED: <Checked Date> DATED: <QC Date> DATED: <Release Date>	COMPANY: <Company Name>			
		TITLE: MT6762				
		CODE: P160AN	SIZE: D	DRAWING NO: P160AN	REV: V1.2	
		SCALE: <Scale>			SHEET: 36 31	

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

D

D

C

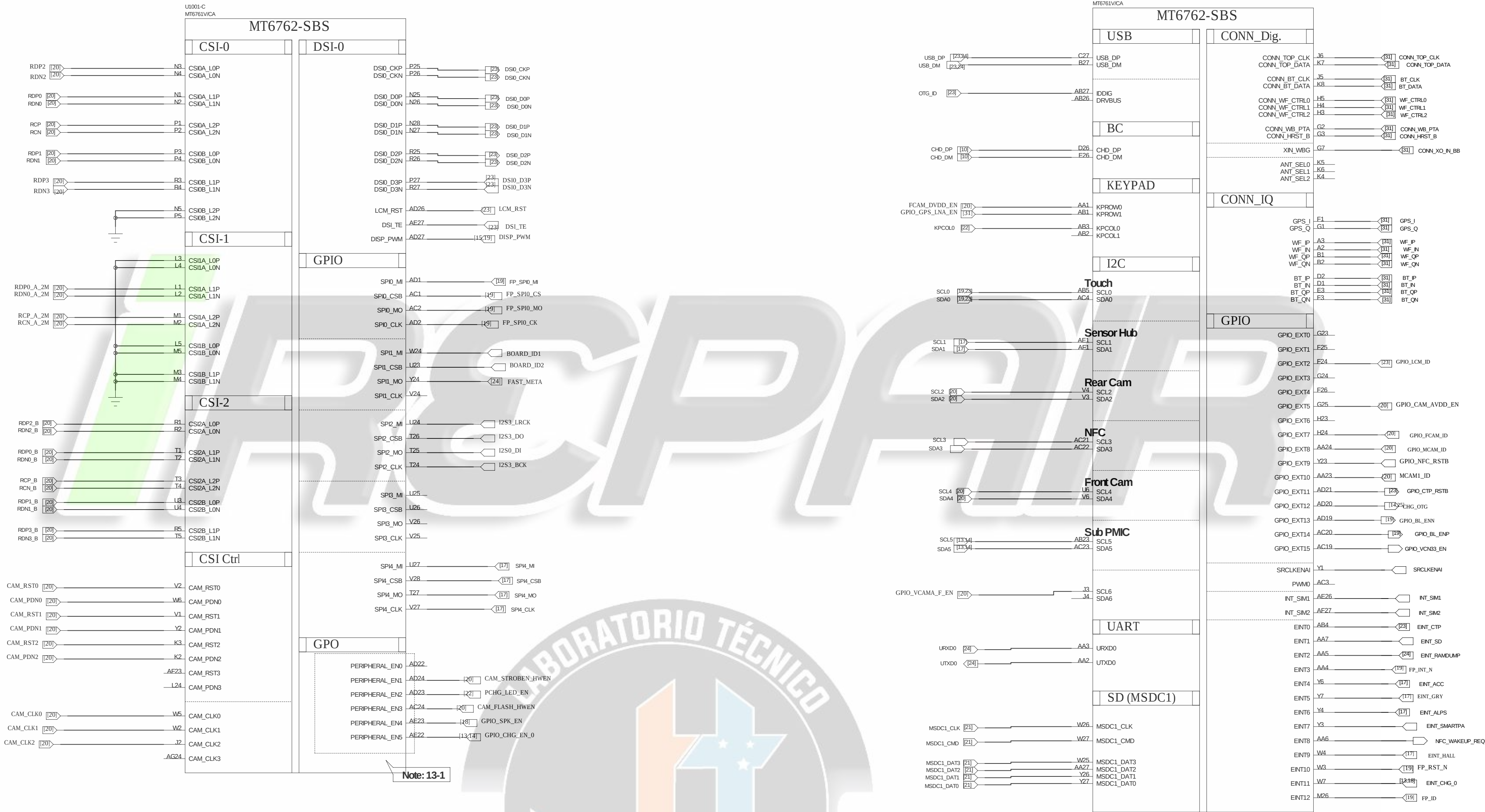
C

B

B

A

A

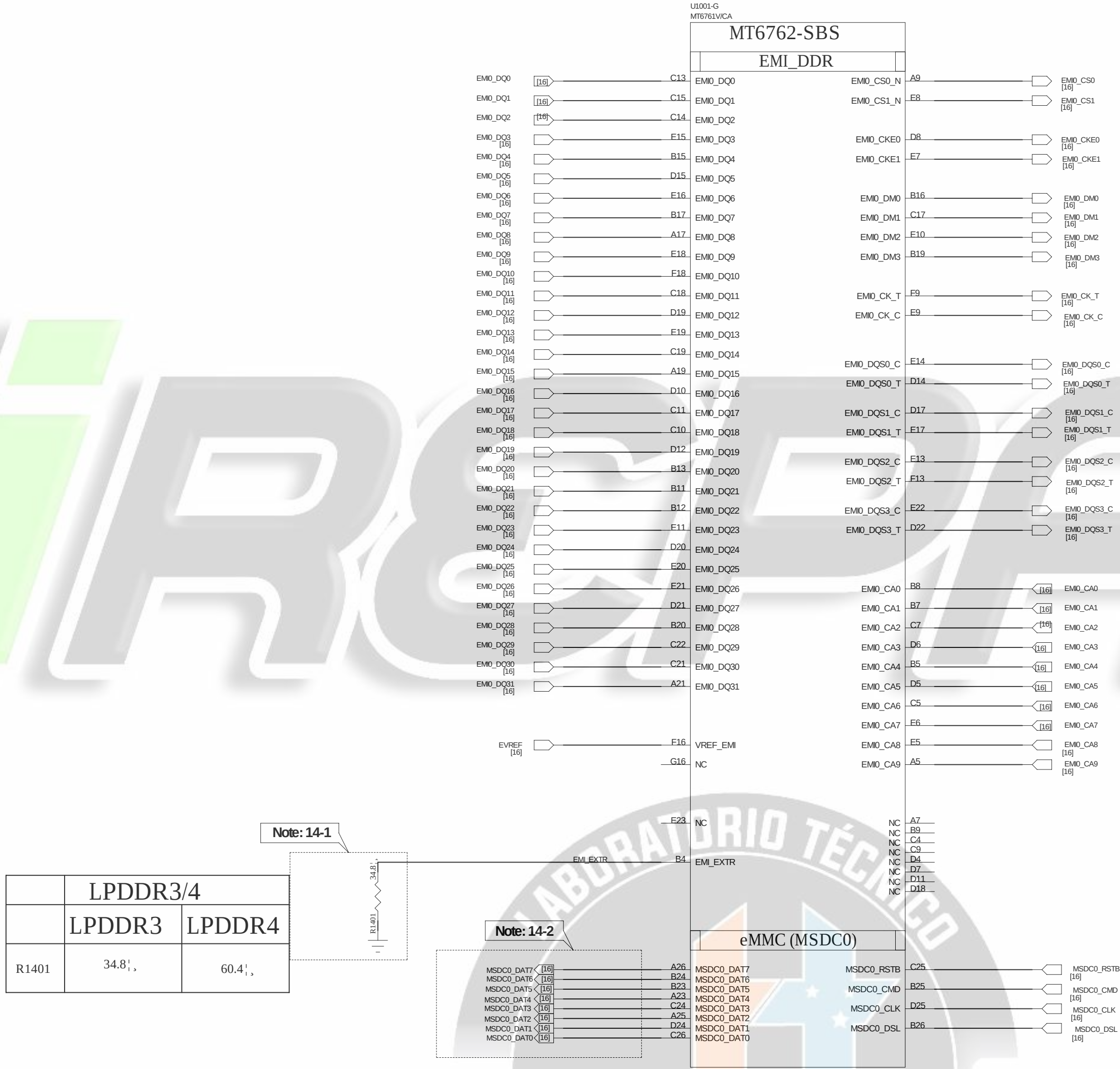


Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

COMPANY: <Company Name>			
TITLE: MT6762			
DRAWN: Chunshen.zhou	DATED: 20170417	CODE: P160AN	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: P160AN	REV: V1.2
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	
RELEASED: <Released By>	DATED: <Release Date>	SHEET 5f 31	

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



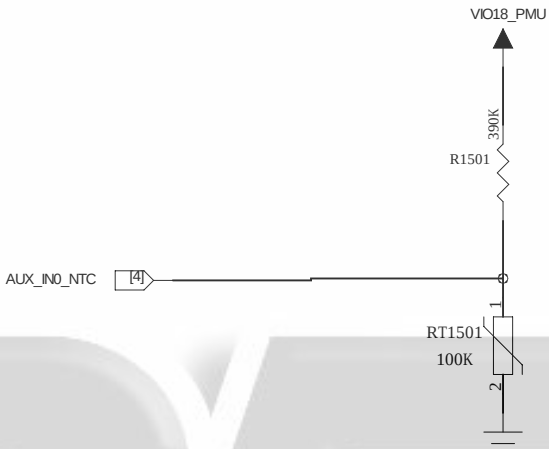
Schematic design notice of "14_BB_3" page.

Note 14-1: R4001 please select 34.8 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux

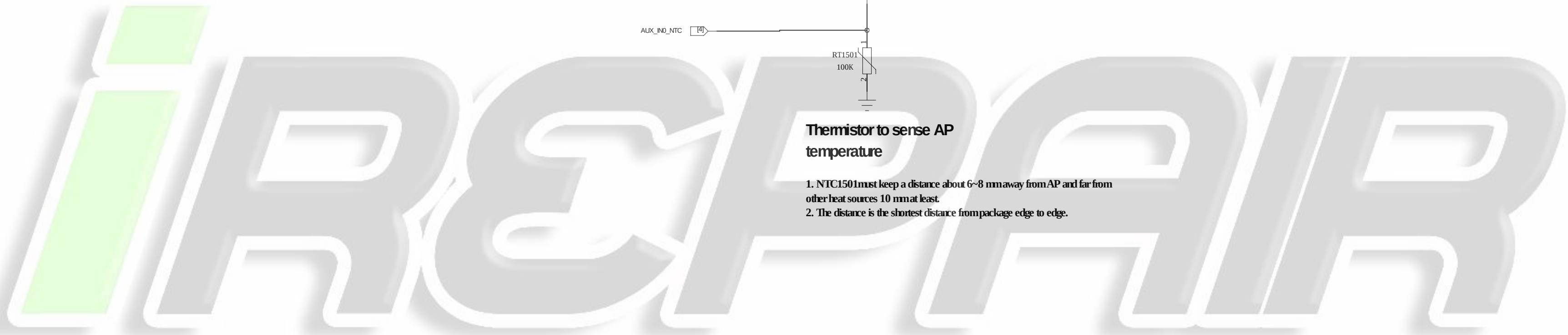
		COMPANY: 			
--	--	--------------	--	--	--

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



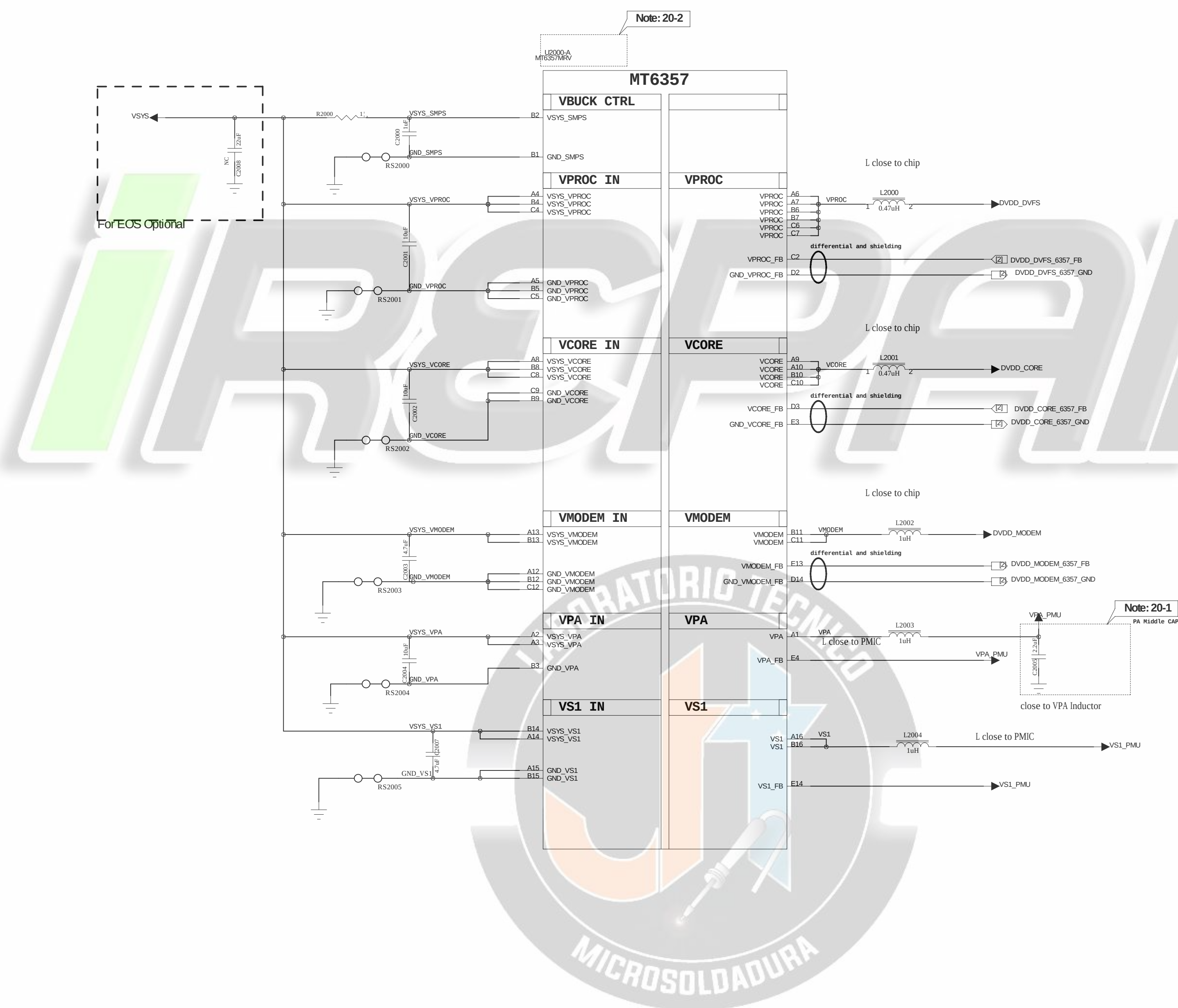
Thermistor to sense AP temperature

- 1. NTC1501 must keep a distance about 6-8 mm away from AP and far from other heat sources 10 mm at least.
- 2. The distance is the shortest distance from package edge to edge.



DRAWN: Chunshen.zhou		DATED: 20170417		COMPANY: <Company Name>			
CHECKED: <Checked By>		DATED: <Checked Date>		TITLE: MT6762			
QUALITY CONTROL: <QC By>		DATED: <QC Date>		CODE: P160AN	SIZE: D	DRAWING NO: P160AN	REV: V1.2
RELEASED: <Released By>		DATED: <Release Date>		SCALE: <Scale>			SHEET: 0f 31

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C2005, please choose 0402 size

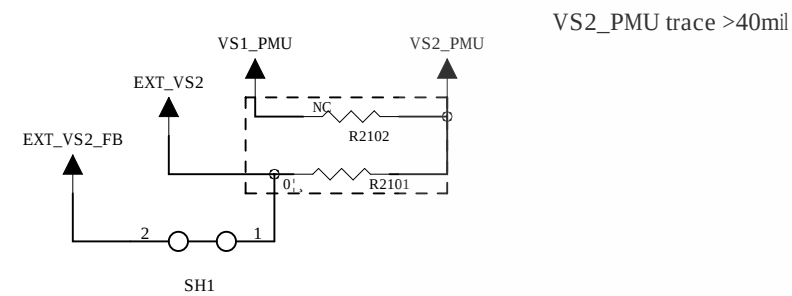
Note 20-2: PMIC Part number notice for MT6765/62/61 platform

MTK Platform	PMIC
MT6765 / 62	MT6357 CRV
MT6761	MT6357 MRV



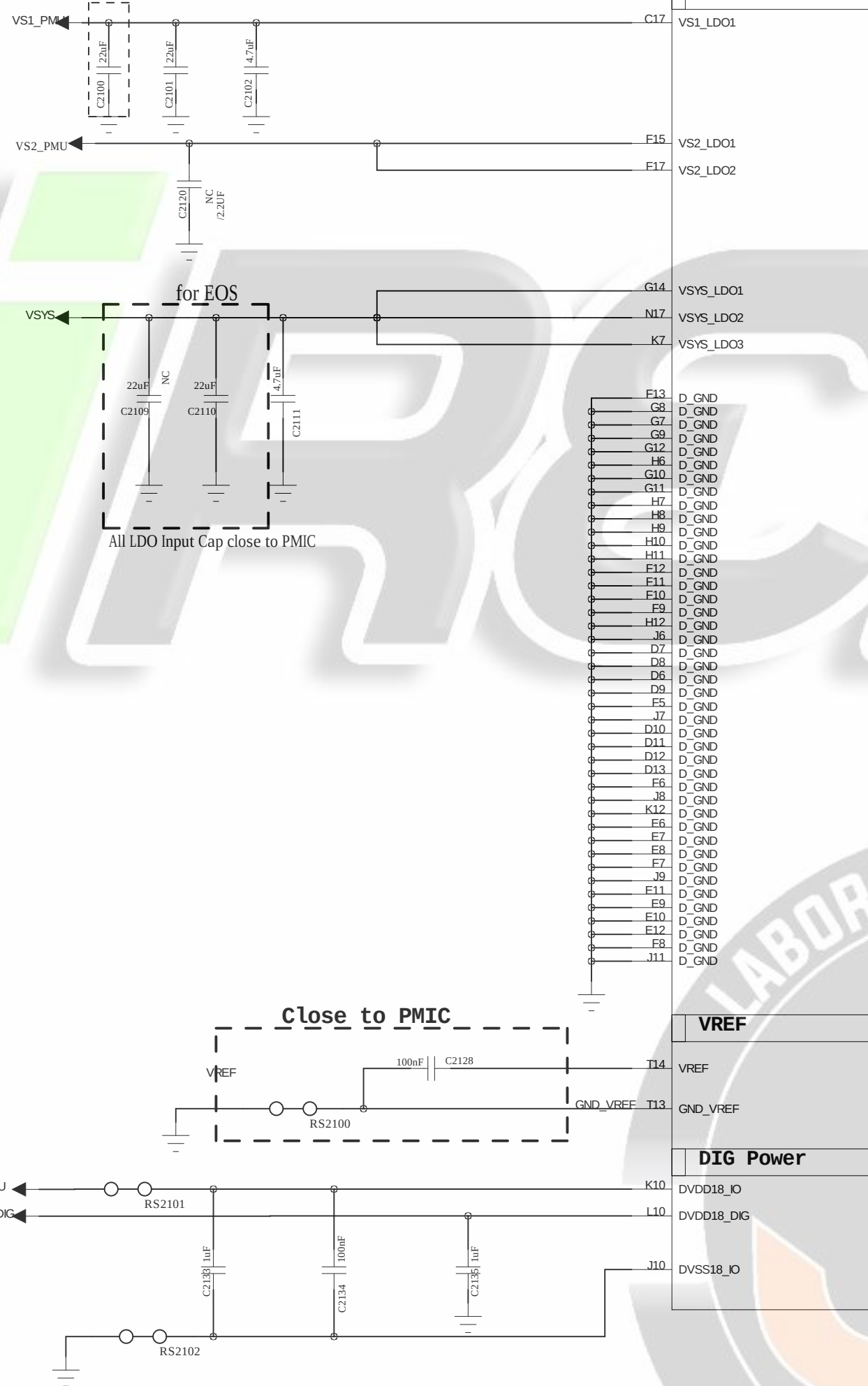
DRAWN: Chunshen.zhou		DATED: 20170417		MT6762	
CHECKED: <Checked By>	DATED: <Checked Date>	CODE: P160AN	SIZE: D	DRAWING NO: P160AN	REV: V1.
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>			SHEET: 8 31

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



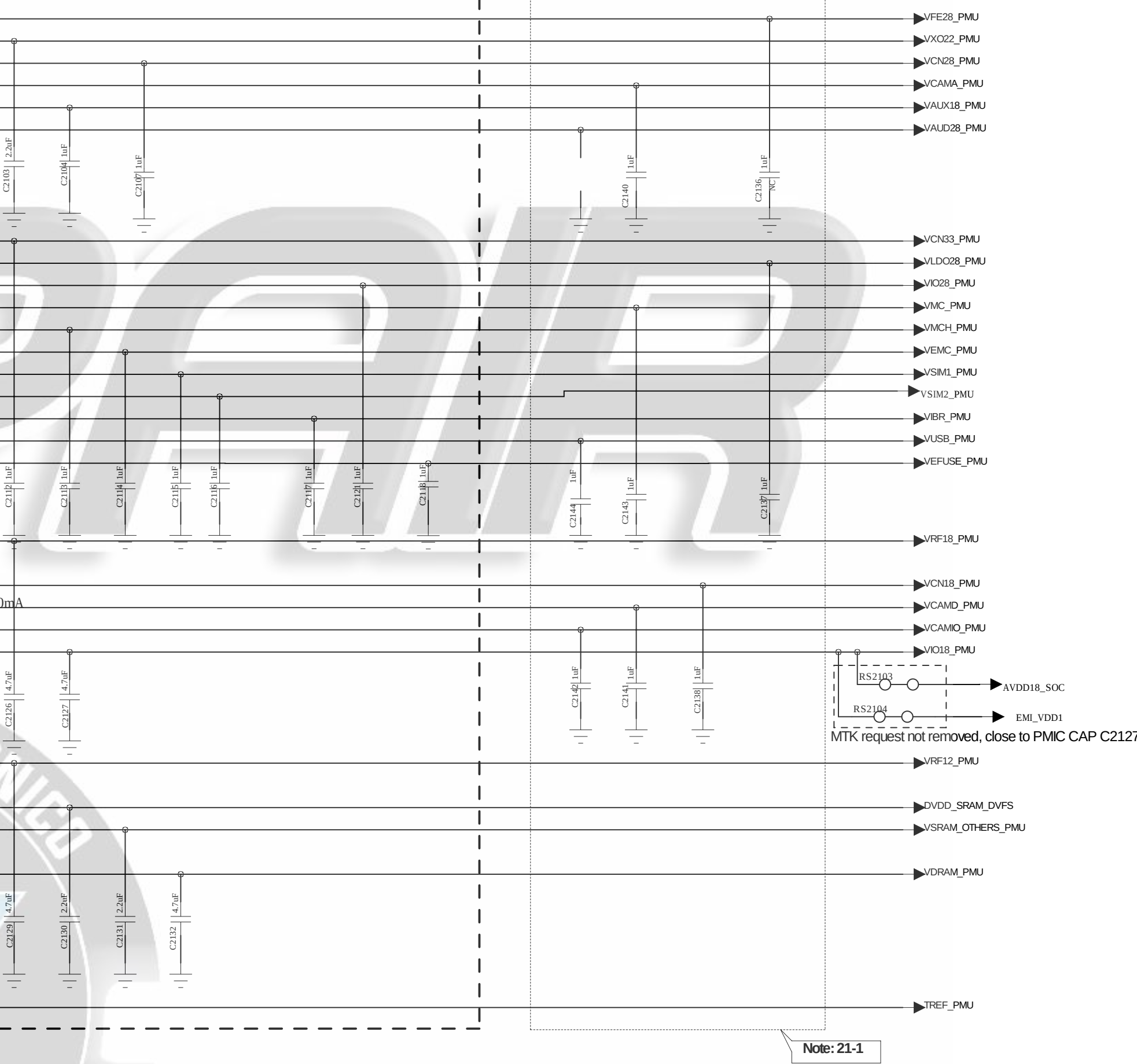
EXIT BUCK OPTION			
	exit VS2 buck-mt6762	W/O exit VS2 buck-mt6762	
C2100	10UF	22UF	
R2101	00HM	00HM	
R2102	NC	00HM	
U2000	MT6357CRV	MT6357MRV	

VS2_PMU trace >40mil



1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
 2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2 ohm)
- => value and placement of Cap, please refer design notice

Close to PMIC



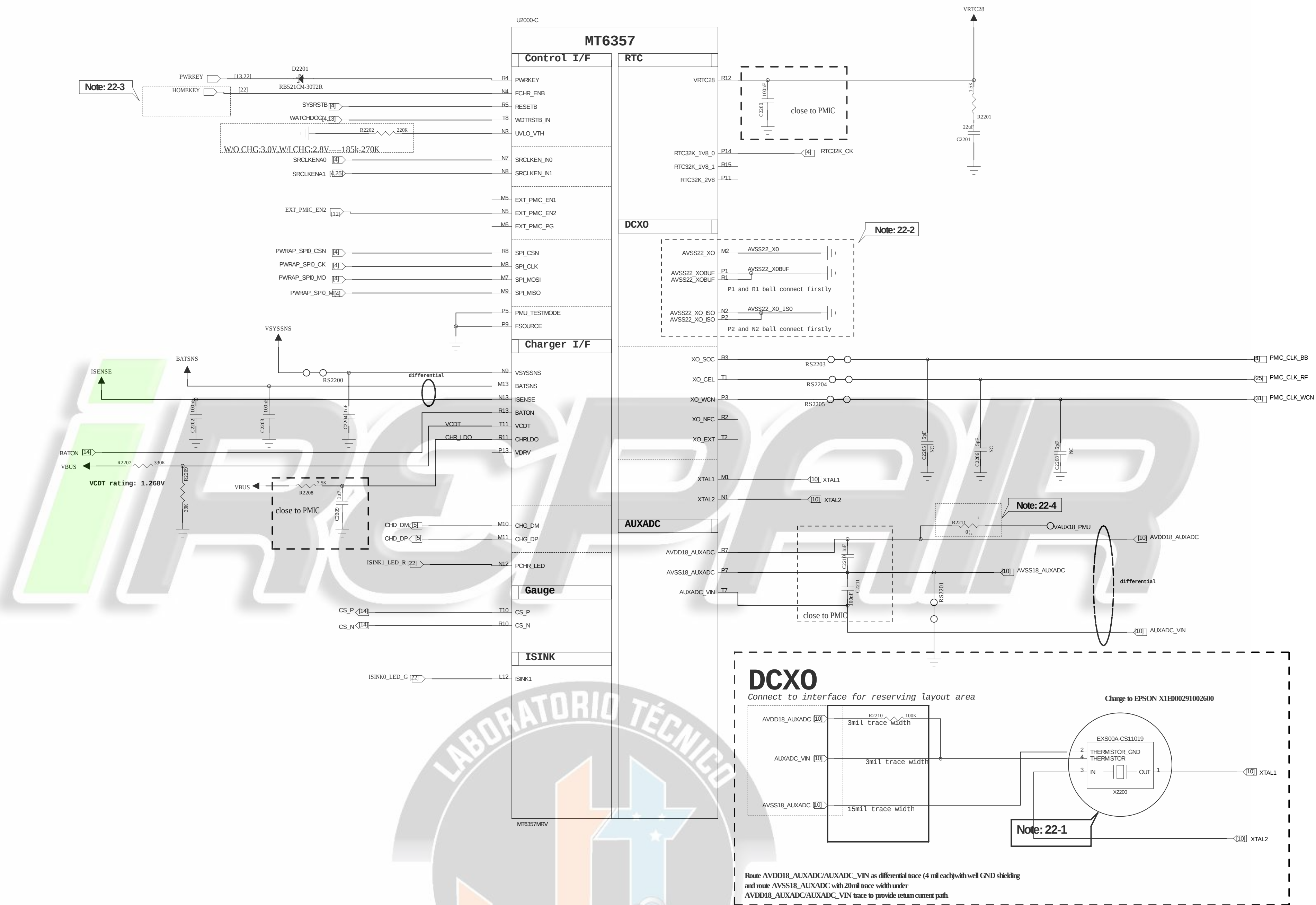
Note: 21-1

Schematic design notice of "21_POWER_MT6357-LDO"

Note 21-1:
If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
Please refer to MT6357 design notice.

COMPANY: <Company Name>			
TITLE: MT6762			
DRAWN: Chunshen.zhou	DATED: 20170417	CODE: P160AN	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: P160AN	REV: V1.2
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	
RELEASED: <Released By>	DATED: <Release Date>	SHEET: 9 of 31	

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "22_POWER_MT6357-IF"

Note 22-1: Please implement 2520 & 2016 Size TMS PCB co-layout.
Please refer to MT6762_MT6357 Co-Clock Design Notice for co-layout guide

Note 22-2:

1. Please Connect P1 and R1 ball first and then to GND
2. Please Connect P2 and N2 ball first and then to GND
3. Please connect DCXO GND to main GND by independent L1-2 GND via.;

Note 22-3: DO NOT connect it through L1 GND
Let floating if disable HOMEKEY function

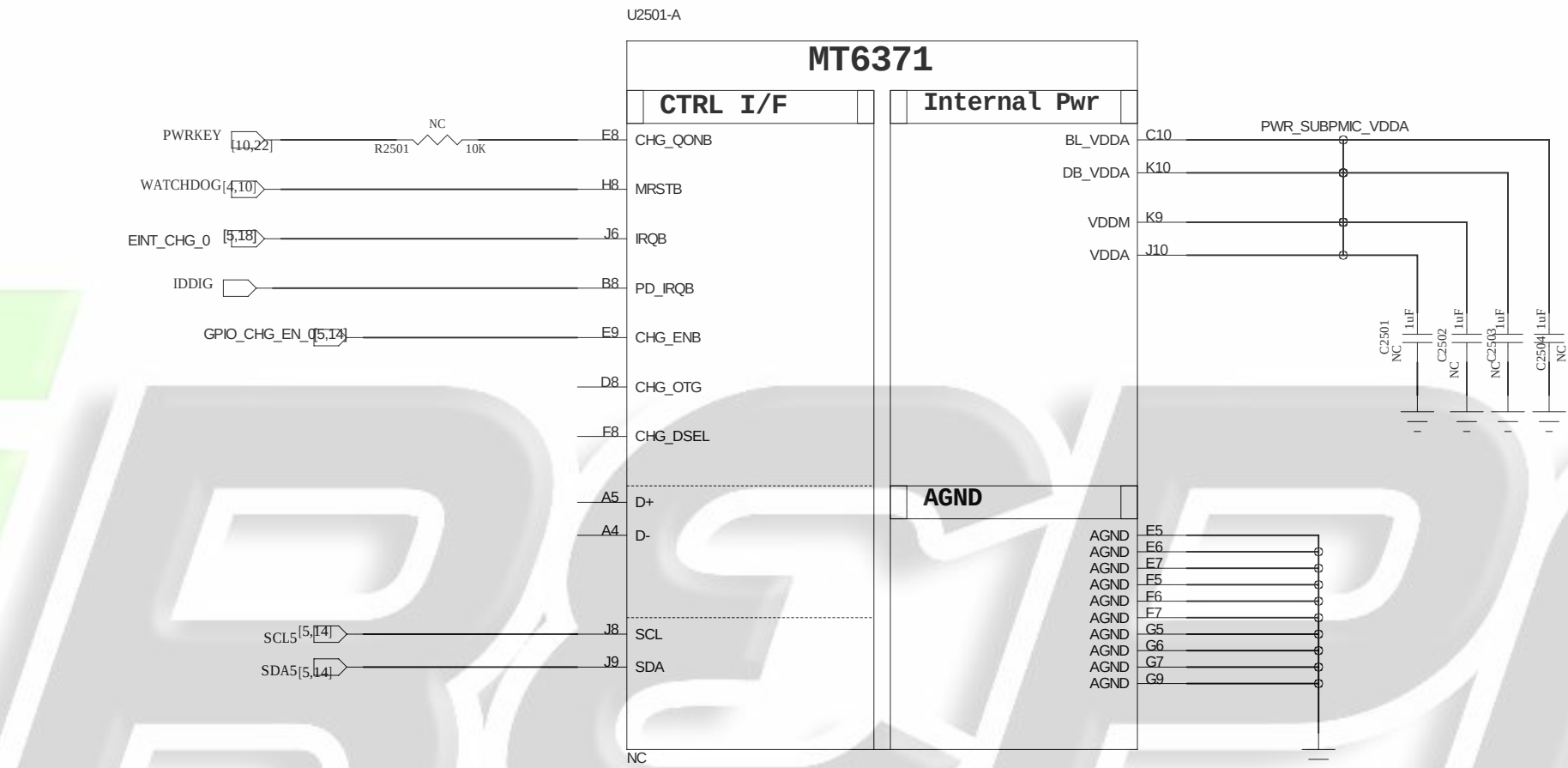
Note 22-4: Please follow MT6762_MT6357 Co-Clock Design Notice for Layout guide of VAUX18, then R2211 can use 0 ohm to replace BEAD.

Note 22-5: Please connect to battery connector

DRAWN: Chunshen.zhou	DATED: 20170417
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY:		<Company Name>	
TITLE:		MT6762	
CODE:	SIZE:	DRAWING NO:	REV:
P160AN	D	P160AN	V1.2
SCALE: <Scale>		SHEET: 00	31

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



DRAWN: Chunshen.zhou		DATED: 20170417		COMPANY: <Company Name>			
CHECKED: <Checked By>		DATED: <Checked Date>		TITLE: MT6762			
QUALITY CONTROL: <QC By>		DATED: <QC Date>		CODE: P160AN	SIZE: D	DRAWING NO: P160AN	REV: V1.2
RELEASED: <Released By>		DATED: <Release Date>		SCALE: <Scale>		SHEET: 48 31	

6

5

4

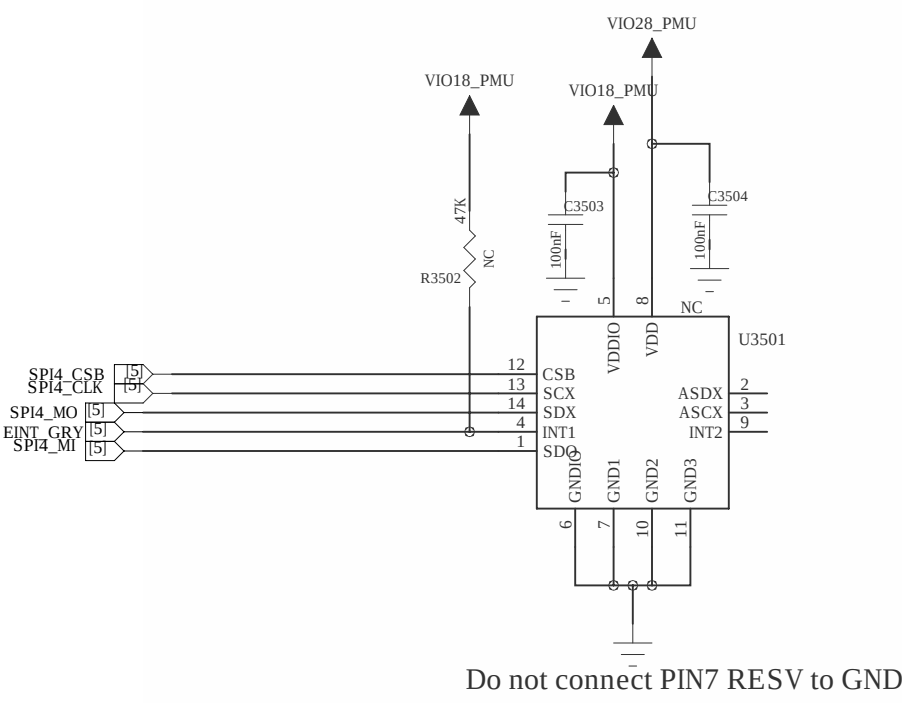
3

2

1

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

Gry-sensor

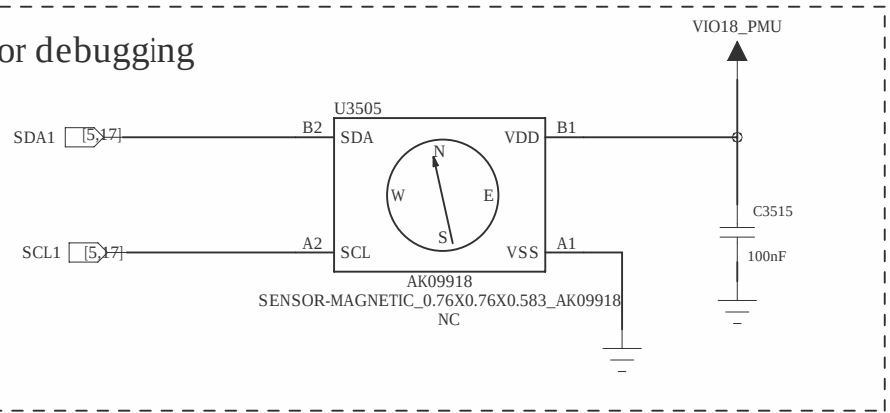


E-Compass

	I2C addr
AF6113	0x0C
AKM09918	0x0C

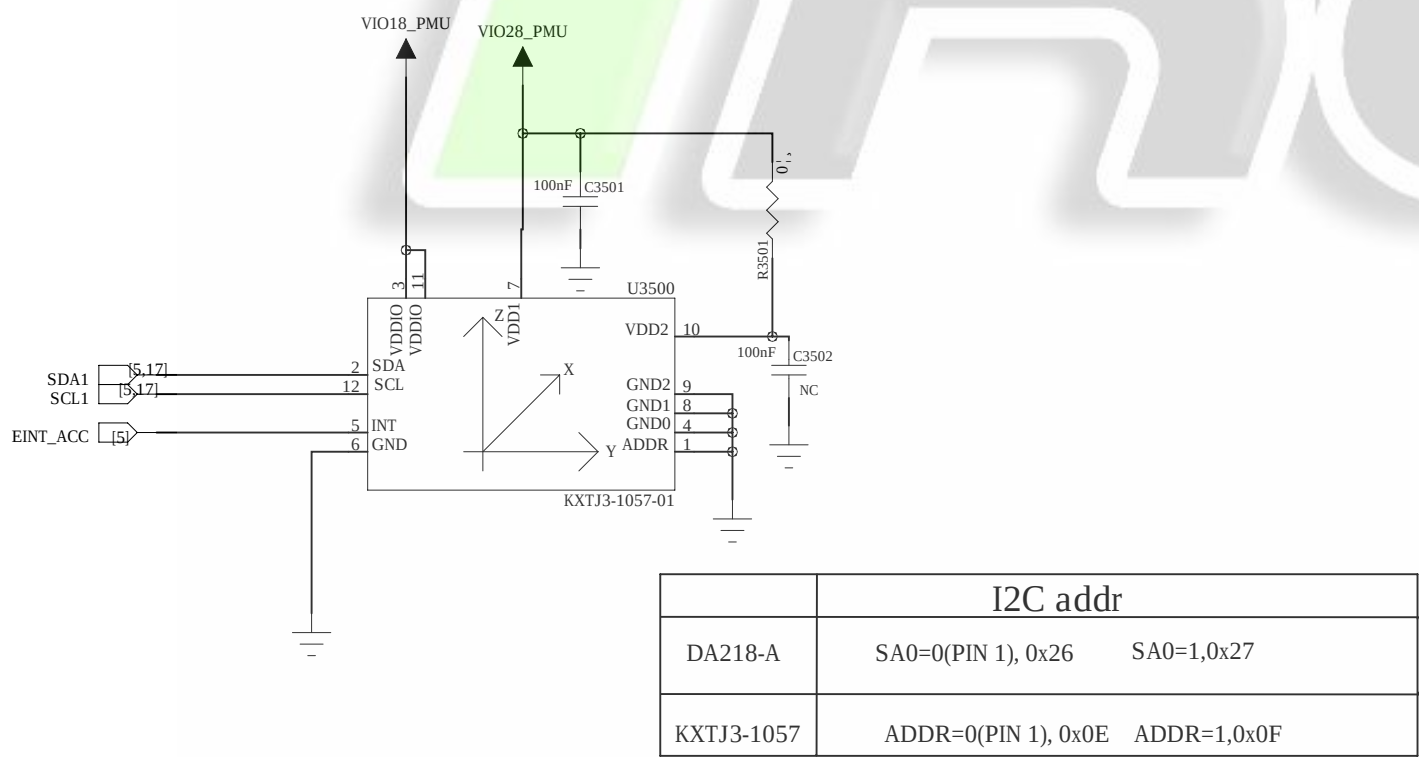
Note: 77-1

For debugging



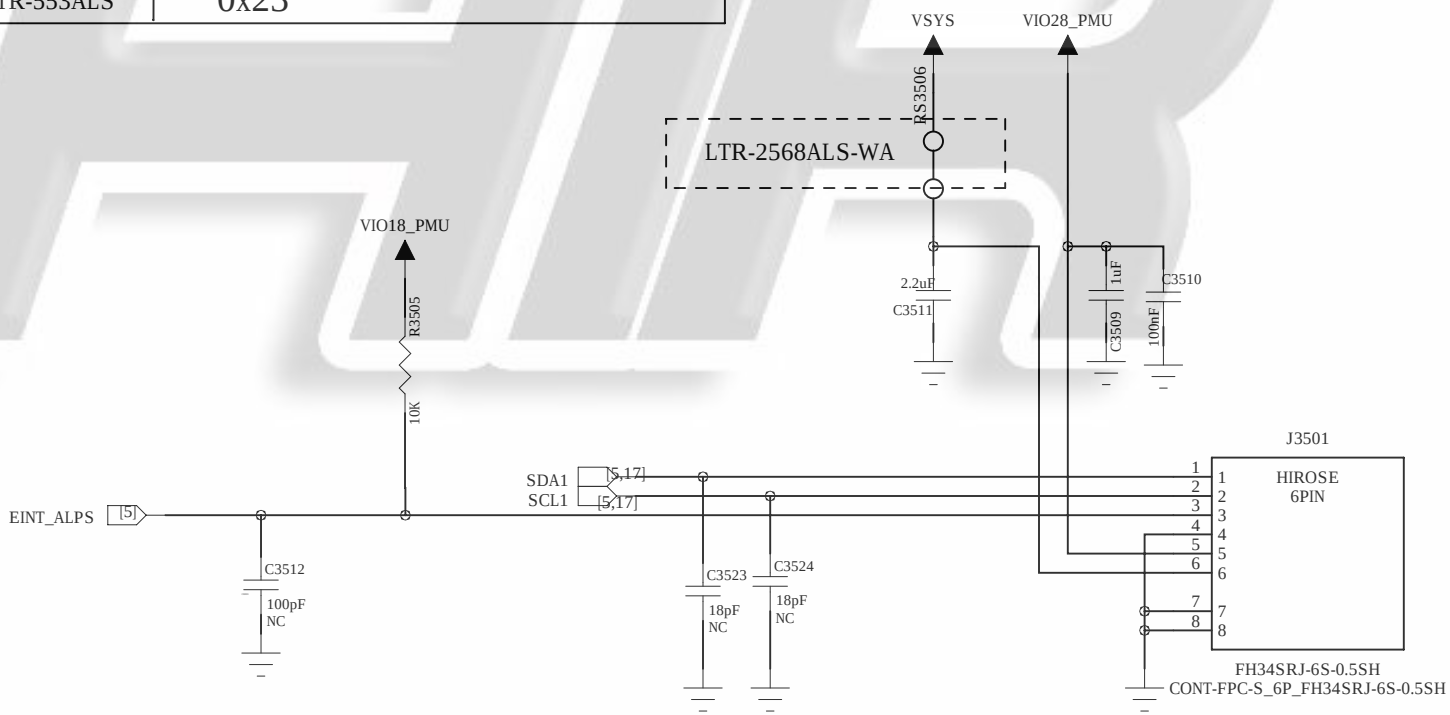
G-sensor

STK8BA50: VDD2 Pin10 is not internally connected
KXTJ3-1057: VDD2 Pin10 is internally connected to VDD pin7. Both decouple with a 0.1uF to GND
DA218-A: VDD2 Pin10 is Digital input, keep floating or connected to 1.8V



Proximity Sensor charge

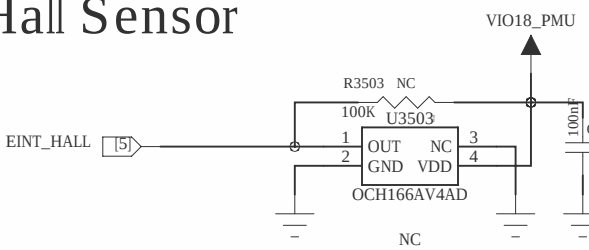
	I2C addr
MN25713EKDN	0x49
LTR-553ALS	0x23



Optional

G+Gyro Sensor

Hall Sensor



Schematic design notice of "77_PERI_SENSORS_MEMs_ALS/PS" page.

Note 77-1: [M sensor] Keep a minimum distance of 15mm from power ICs / PCB traces of more than 100mA / magnet component. Check HW design notice for more detail

Note 77-2: [A+G] For optimized GPS performance, please check HW design notice for Sensor selection guide

Note 77-3: [A+G] MUST use SPI for optimized sensor hub performance DO NOT USE I2C

Note 77-4: [A+G] Suggest choose sensor support FIFO watermark interrupt otherwise we cannot support Hifi-sensor, daydream VR. And Sensor-location accuracy will become worse.

Note 77-5: [Baro] Reserve Baro sensor for LPPe feautre (Must for North America Operator / NA SKU)

Note 77-6: DO NOT share Sensor hub i2C to other non-SCP device

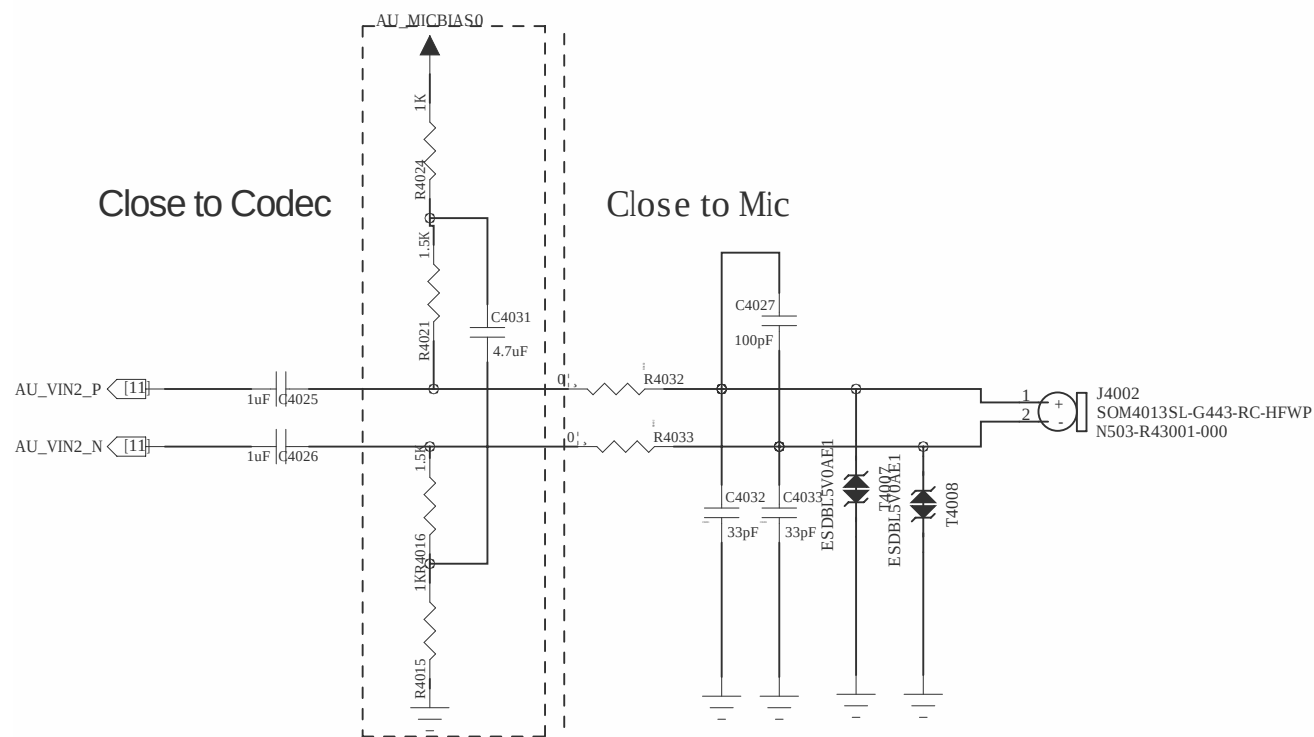
Note 77-7: Interrupt pin of MEMS sensor must be assign to ball EINT[12:0]

COMPANY: <Company Name>			
TITLE: MT6762			
DRAWN: Chunshen.zhou	DATED: 20170417	CODE: P160AN	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: P160AN	REV: V1.2
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	SHEET: 31
RELEASED: <Released By>	DATED: <Release Date>		

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

Sub Mic

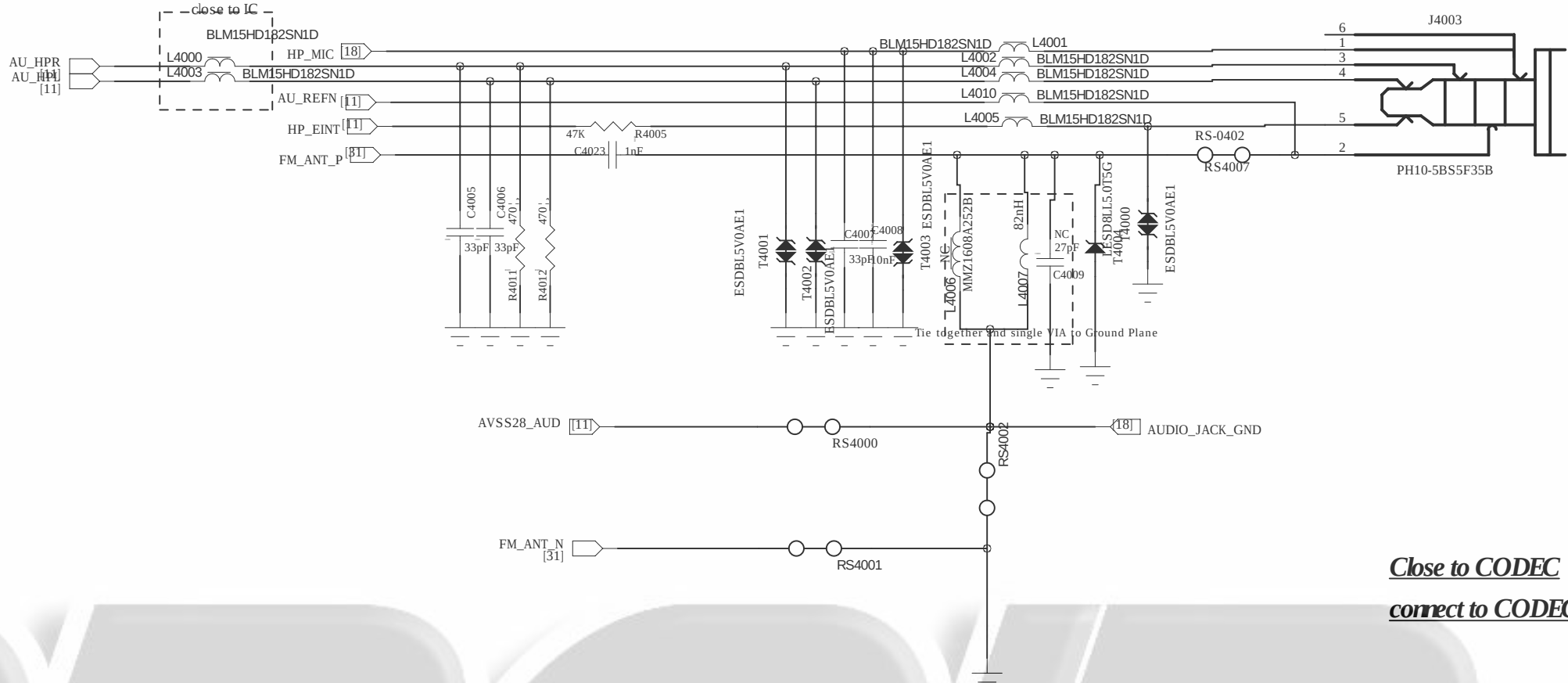
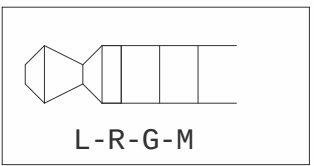
use ACC mode



Close to Codec

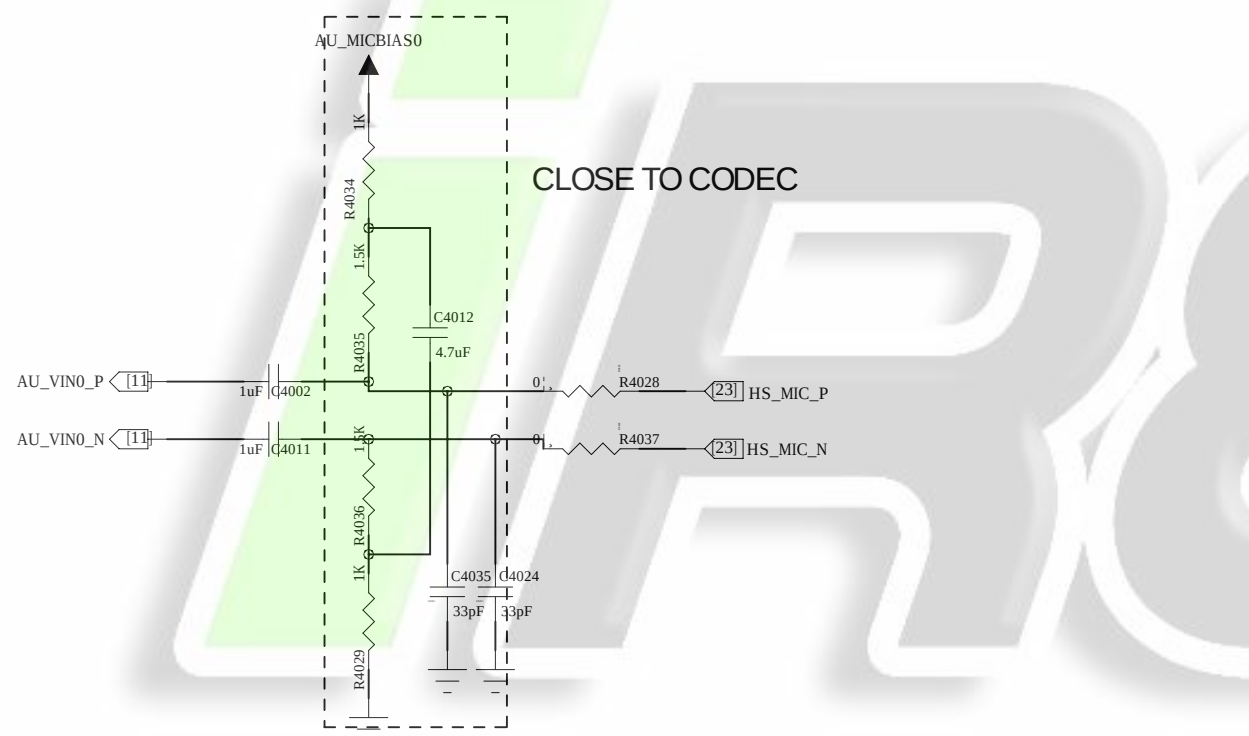
Close to Mic

CTIA Mode



Main Mic

use ACC mode

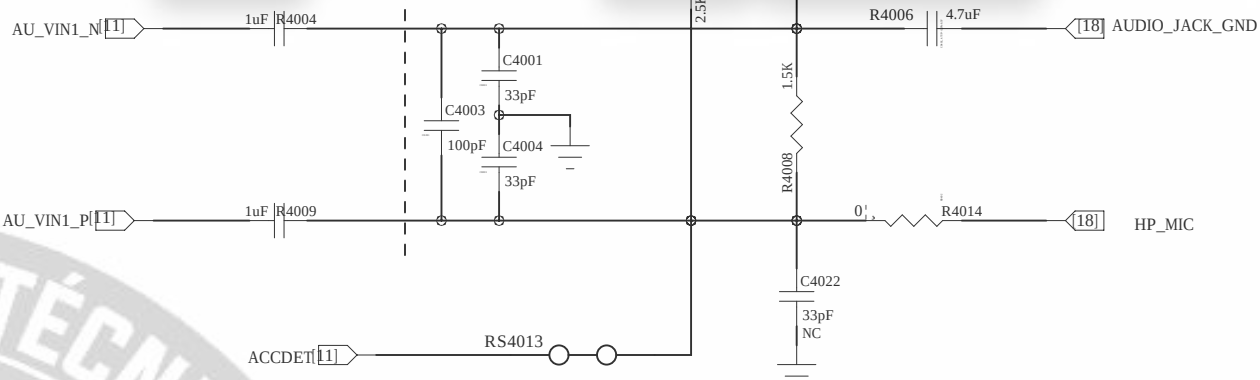


CLOSE TO CODEC

Headset Mic

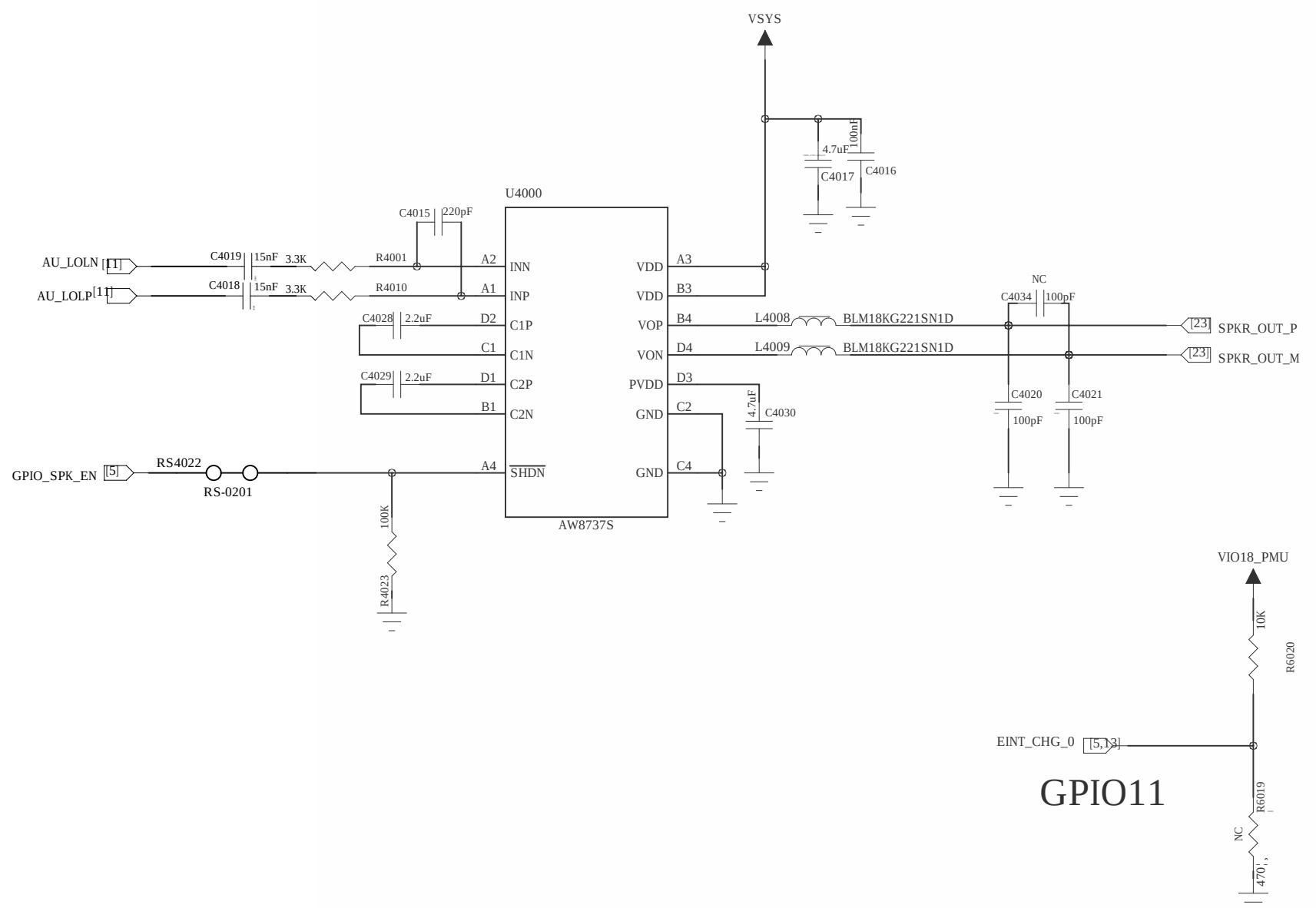
ACC mode

Close to Codec



	DCC	ACC
R4004	0R	1uF
R4009	0R	1uF
R4003	NC	1K
R4008	NC	1.5K
R4002	2.5K	NC
R4006	0R	4.7uF
R4013	NC	0R

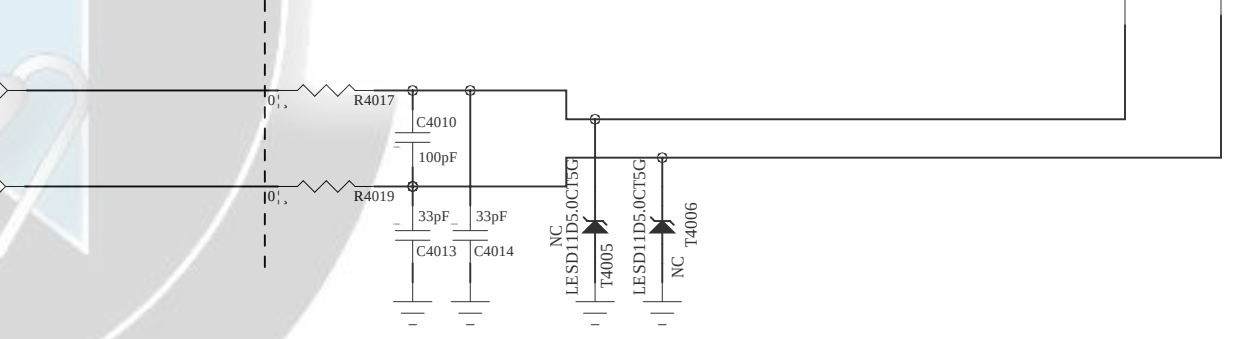
SPK AMP



Handset Receiver

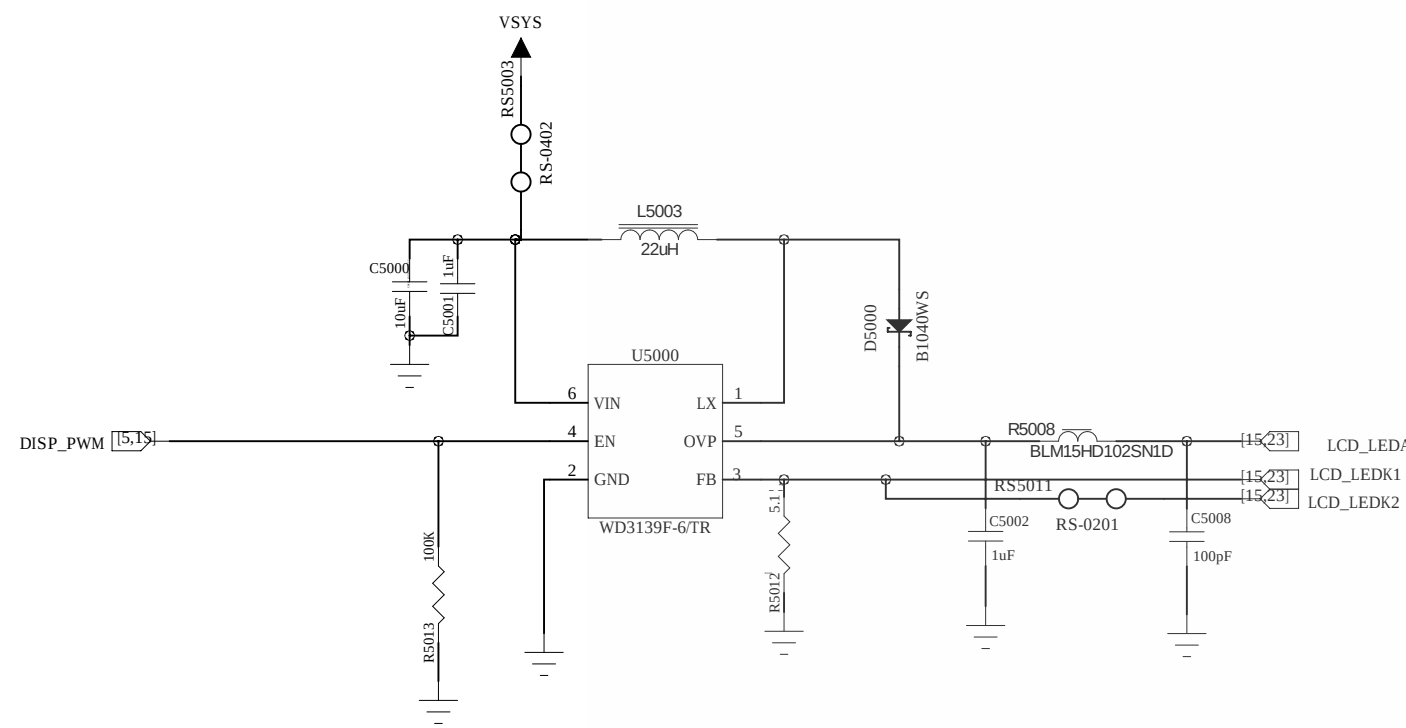
Close to Codec

Close to Receiver



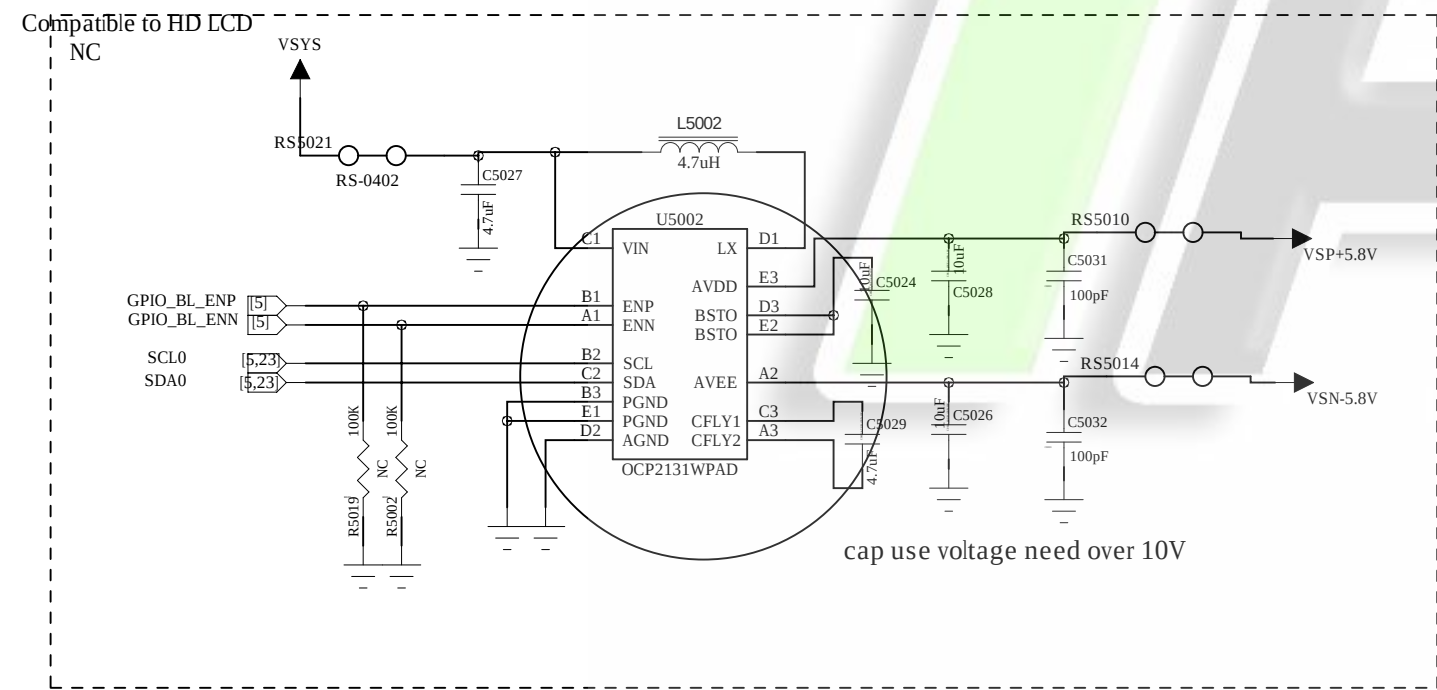
COMPANY: <Company Name>			
TITLE: MT6762			
DRAWN: Chunshen.zhou	DATED: 20170417		
CHECKED: <Checked By>	DATED: <Checked Date>	CODE: P160AN	SIZE: D
QUALITY CONTROL: <QC By>	DATED: <QC Date>	DRAWING NO: P160AN	
RELEASED: <Released By>	DATED: <Release Date>	REV: V1.2	
SCALE: <Scale>		SHEET: 48 31	

LCD Backlight



WD3139F is P2P Compatible to SGM3752

LCD BIAS

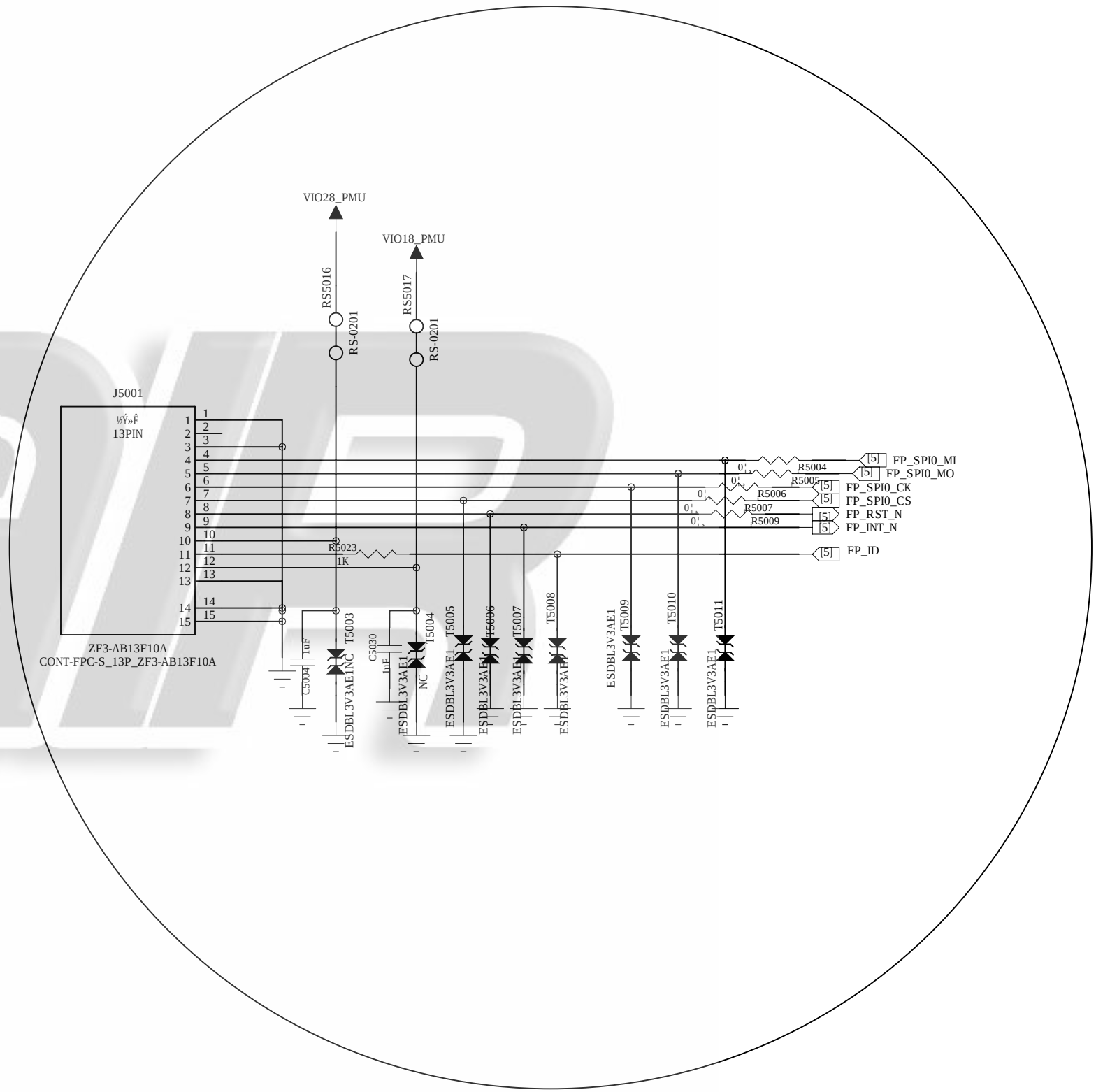


LCD Connector

Compatible to HD LCD

Fingerprint Connector

1	GND
2	HOME(NC)
3	GND
4	MISO
5	MOSI
6	CLK
7	CS
8	RST
9	INT
10	VDD2V8
11	ID(GND)
12	DOVDD1V8
13	GND

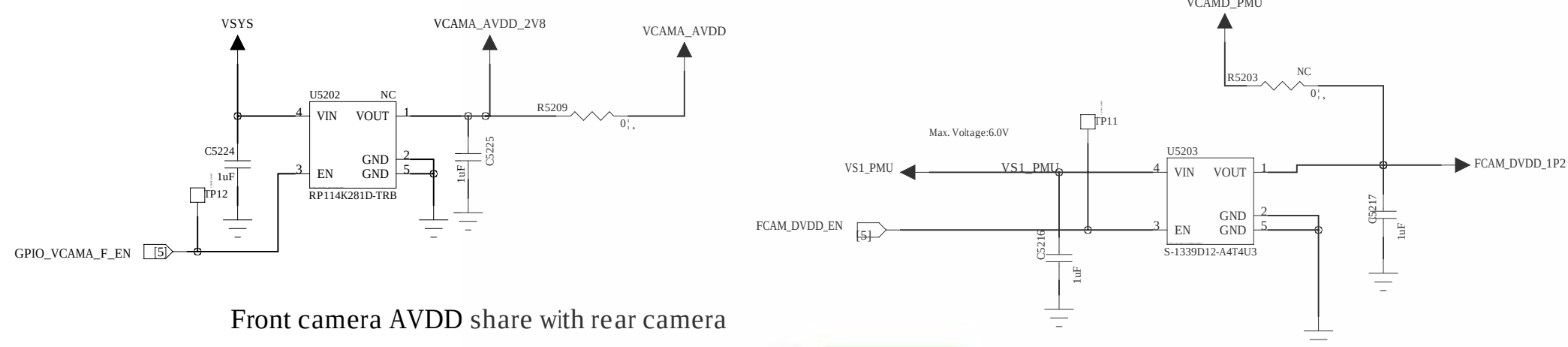


COMPANY: <Company Name>				
TITLE: MT6762				
DRAWN: Chunshen.zhou	DATED: 20170417	CODE: P160AN	SIZE: D	DRAWING NO: P160AN
CHECKED: <Checked By>	DATED: <Checked Date>	QUALITY CONTROL: <QC By>	DATED: <QC Date>	REV: V1.2
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>		

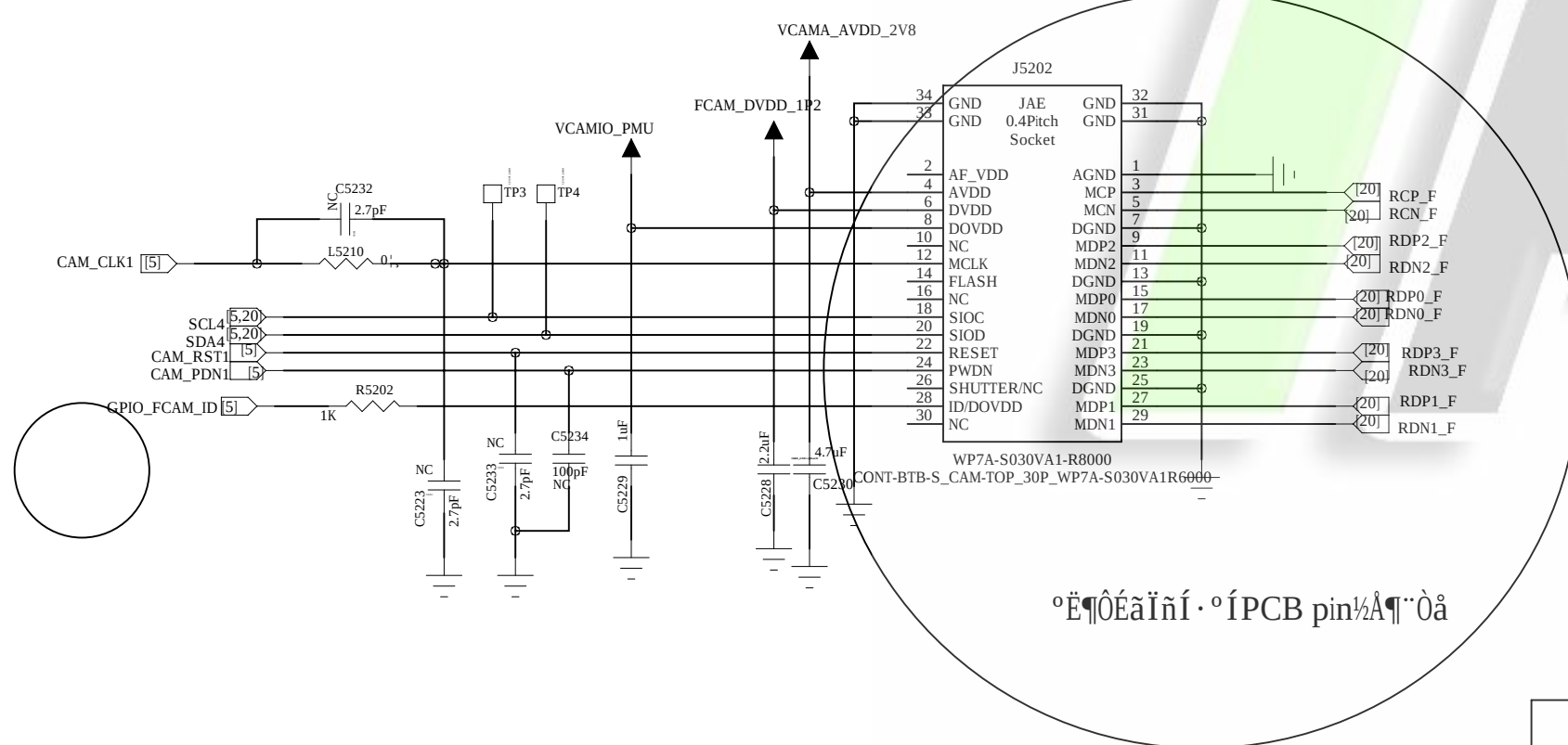
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

	AVDD	DVDD	DOVDD	ID	ADDR
GC5025(cameraking)	2.7~3.0V/20ma	1.15~1.25V/60ma	1.7~1.9V/2ma	GND	0X37
GC5025(SunWin)	2.7~3.0V/20ma	1.15~1.25V/60ma	1.7~1.9V/2ma	DOVDD	0X37
OV5670(SunWin)	2.7~3.0V/20ma	1.15~1.25V/60ma	1.7~1.9V/2ma	DOVDD	0X36
GC030A(cameraking)	2.8V	NC	1.8V	GND	0X42/0X43(R)

	AVDD	DVDD	DOVDD	ID	I2C ADDR(7bits)	VCN:V8B860I	AF_VDD
S5K3L6XX SAMSUNG	2.8V	1.0.5V	1.8V		0X21H(R);0X20H(W)		2.8V
sp2509(cameraking)	2.8V	NC	1.8V			NC	NC

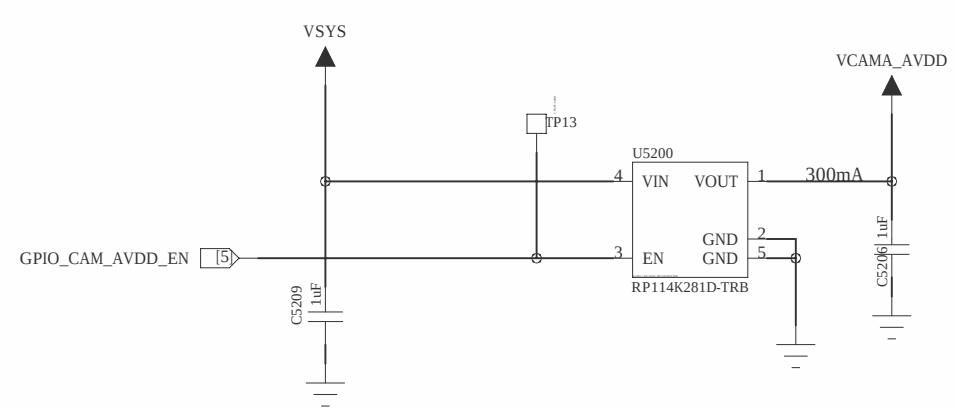


Front-CAMERA

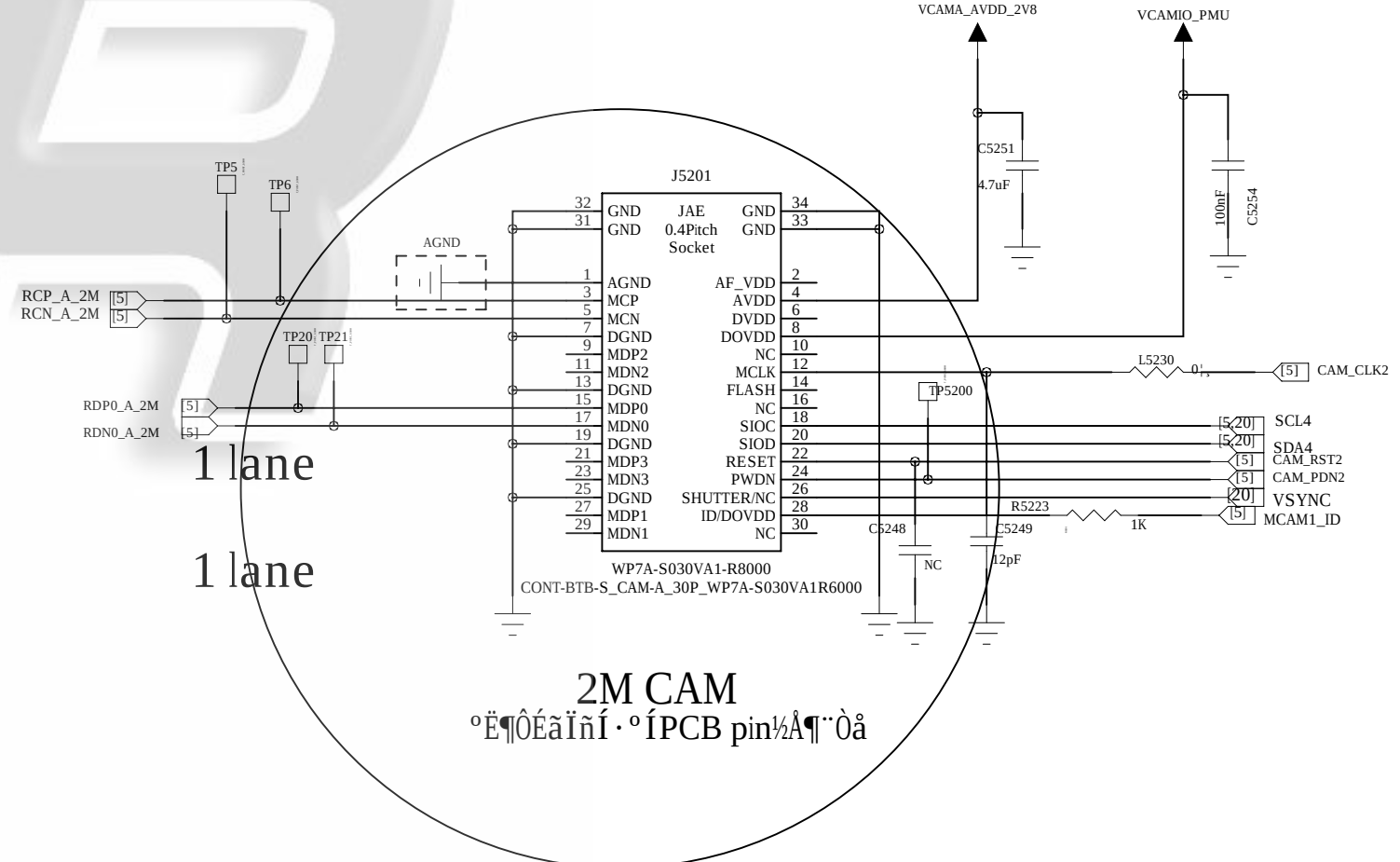


	8M/5M CAM	
	5M CAM/16+2	8M CAM/32+2
R5230	0 _i ,	NC
R5231	0 _i ,	NC
R5232	0 _i ,	NC
R5233	0 _i ,	NC

RF requested caps for mcam desense ,close to connecto



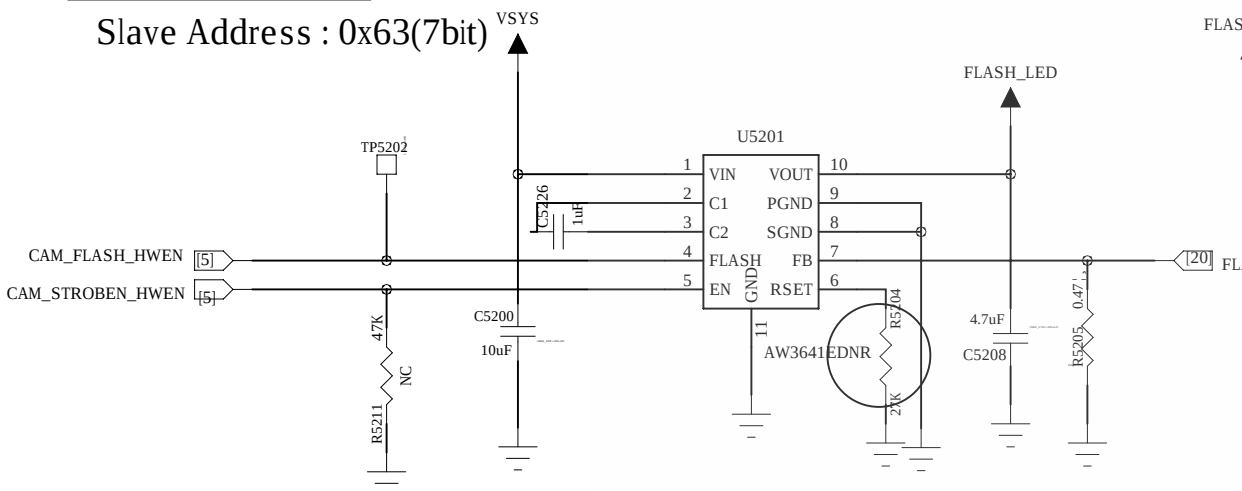
Depth-camera
power on sequence : IOVDD > AVDD



E5201,E5202ĐÍ°Å´ýÈ·Èİ



Slave Address : 0x63(7bit) ^{VS}



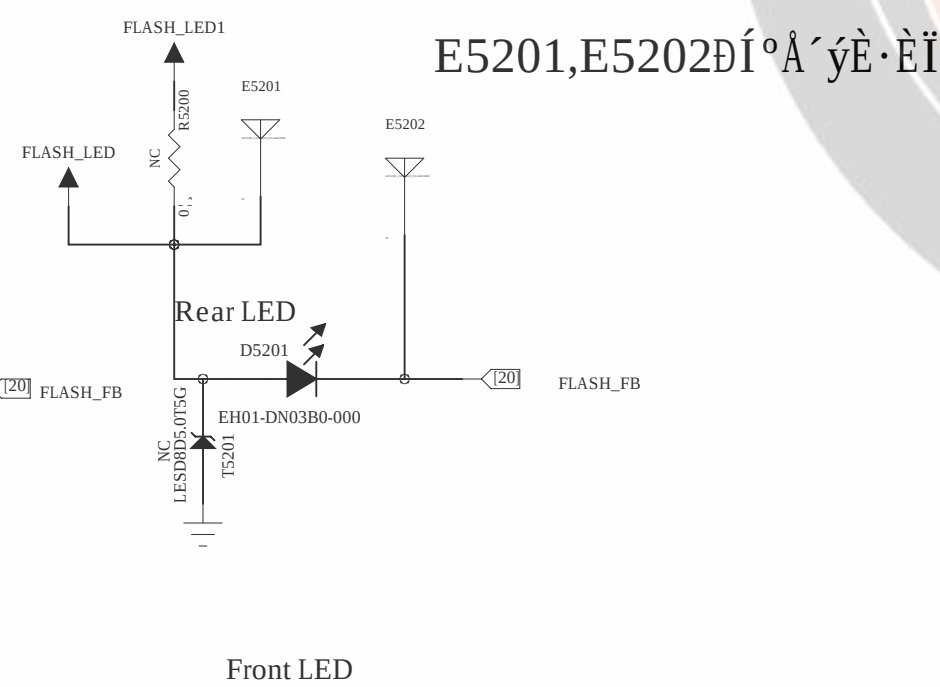
OCP8110 R5204=82K

AW3641 R5204=56K

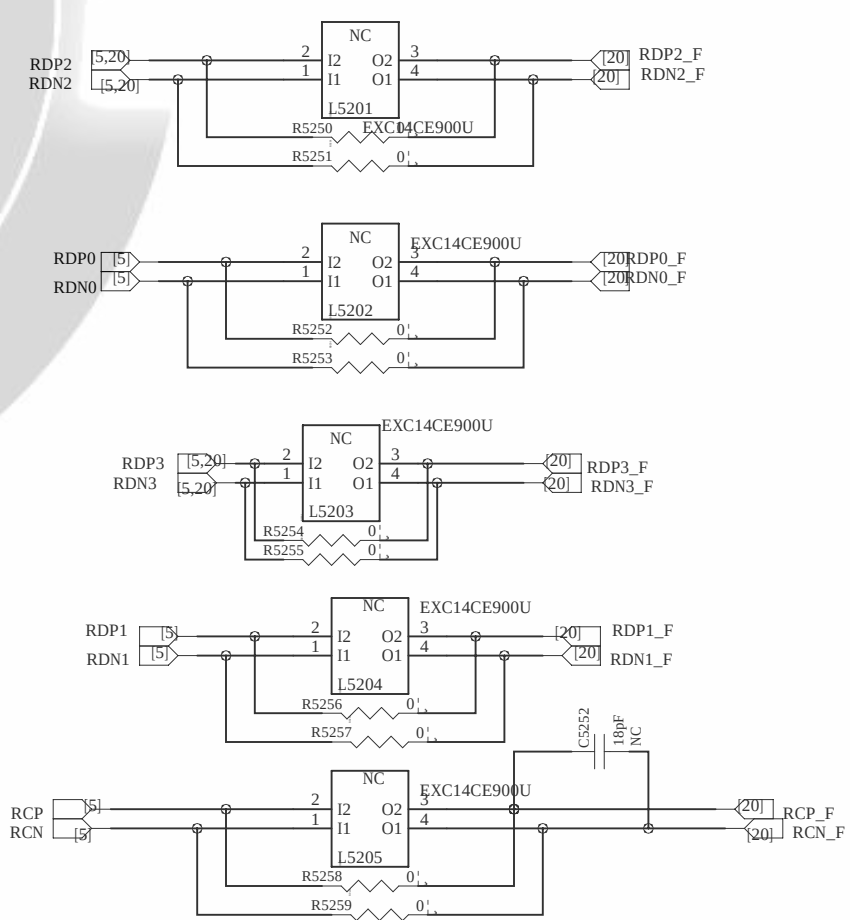
AW3641 R5204=27K

Torch current= $50\text{mV}/R_{5205}=106\text{mA}$
Flash current= $(14\mu\text{A}\cdot R_{5204})/(5\cdot R_{5205})=489\text{mA}$

Torch current= $47\text{mV}/R_{5205}=100\text{mA}$
Flash current= $(1.26\text{V}/R_{5204}/R_{5205})\times 10.2\text{K}=488\text{mA}$

$$\text{Flash current} = (1.26\text{V}/R5204/R5205) * 10.2\text{K} = 1.01\text{A}$$


Front LED



DRAWN: Chunshen.zhou	DATED: 20170417
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY:		<Company Name>	
TITLE:			
MT6762			
CODE:	SIZE:	DRAWING NO:	REV:
P160AN	D	P160AN	V1.2
SCALE: <Scale>		SHEET: 30	31

6

5

4

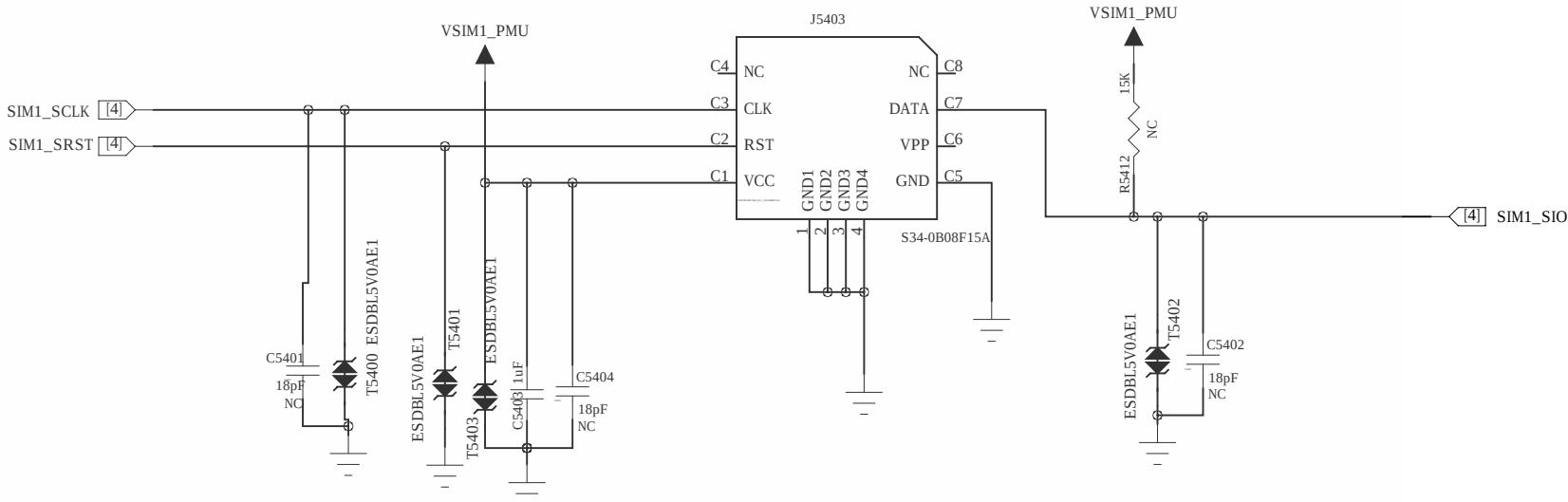
3

2

1

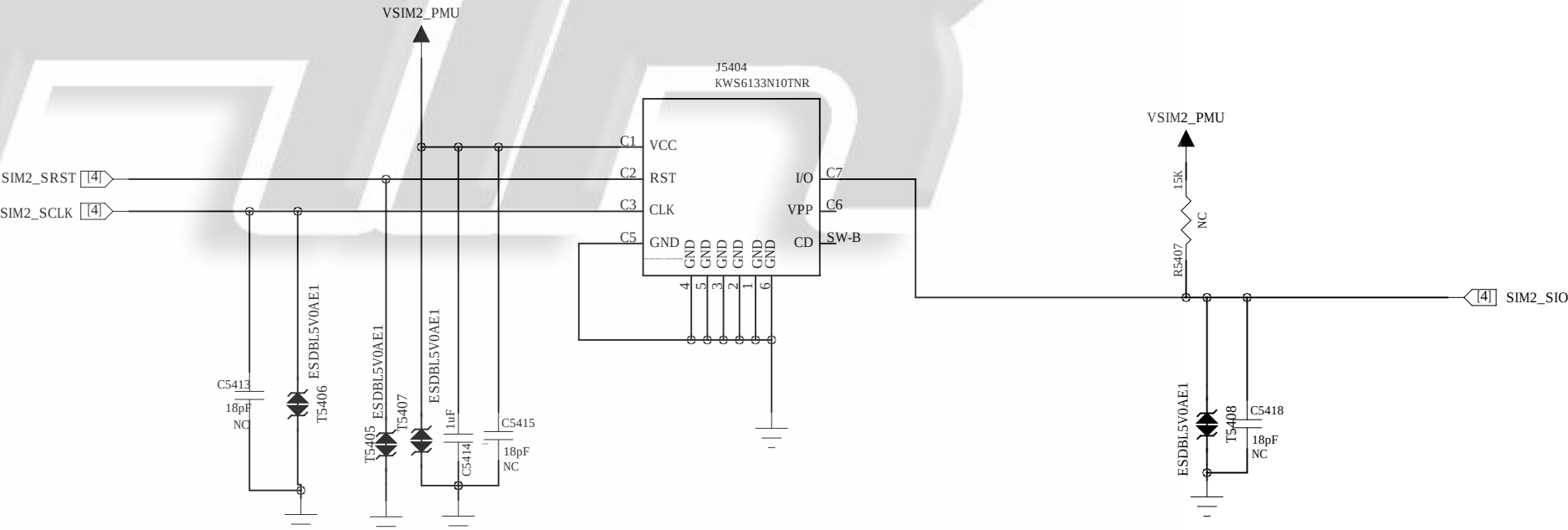
SIM1

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

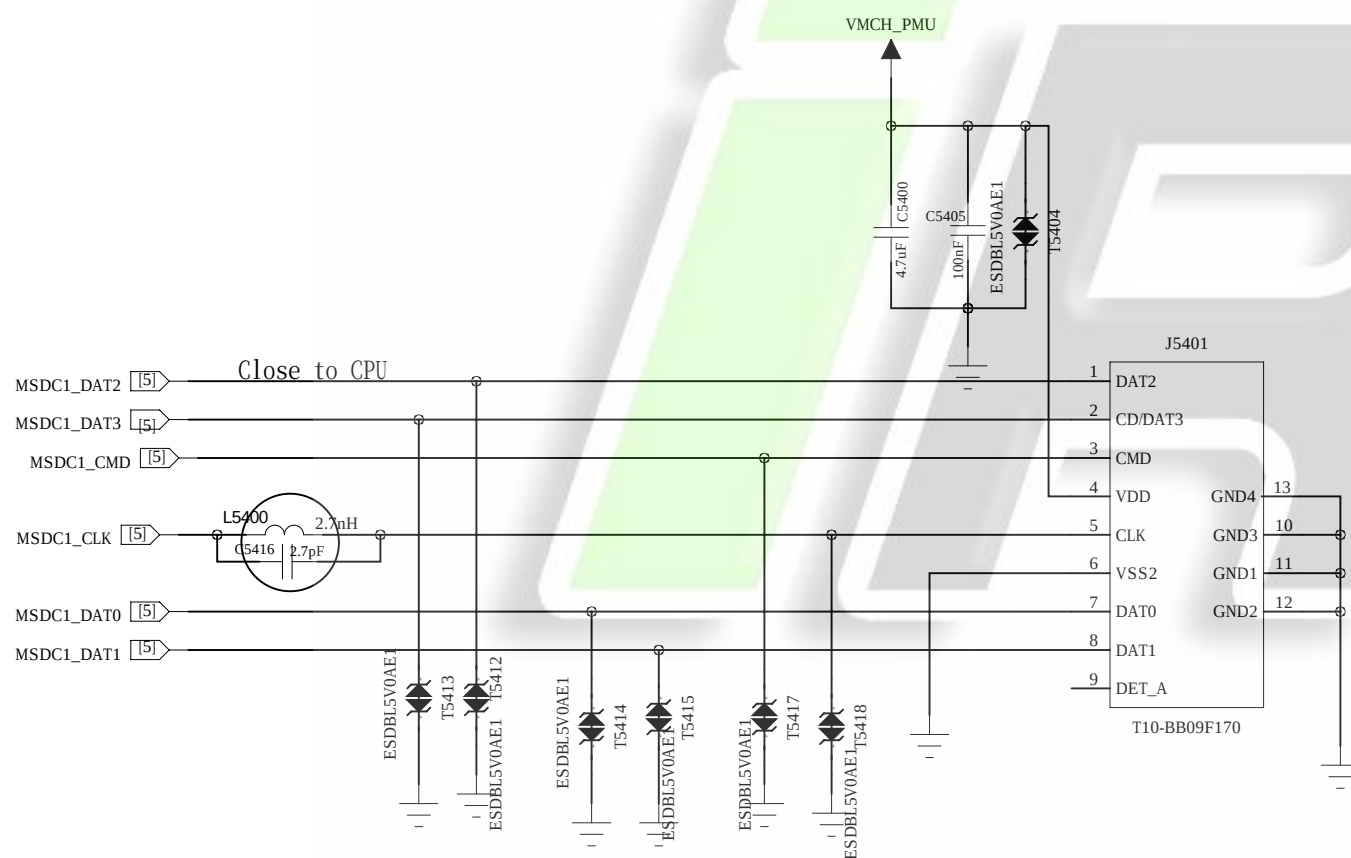


Normal:0
Insert:1

SIM2



TF Card



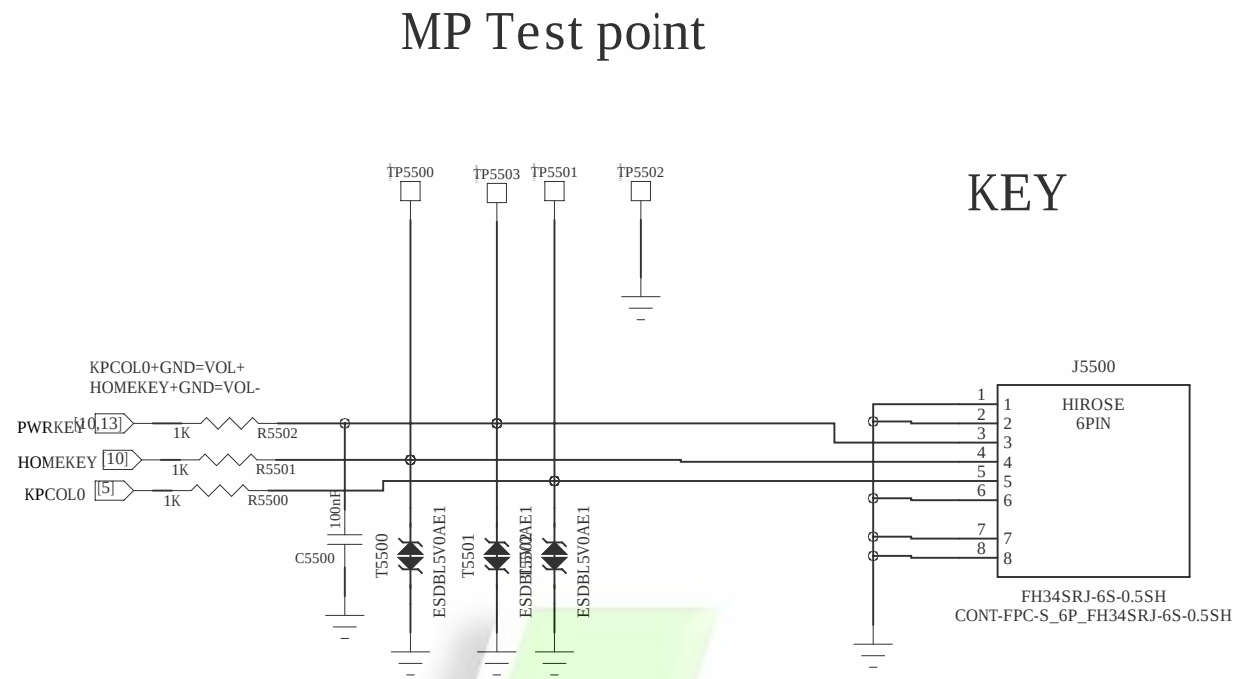
Normal:0
Insert:1



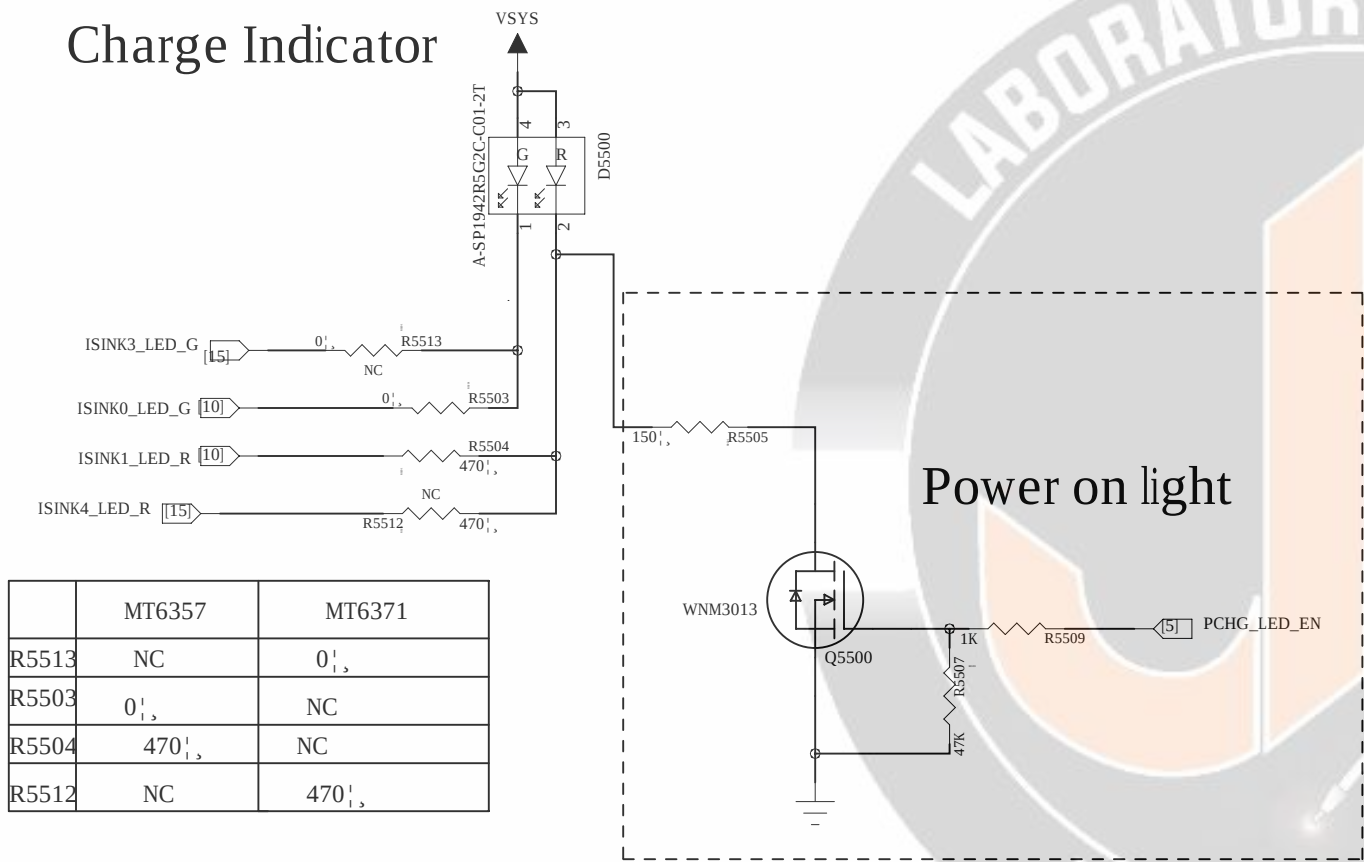
DRAWN: Chunshen.zhou	DATED: 20170417
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY: <Company Name>			
TITLE: MT6762			
CODE: P160AN	SIZE: D	DRAWING NO: P160AN	REV: V1.2
SCALE: <Scale>		SHEET 31 31	

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

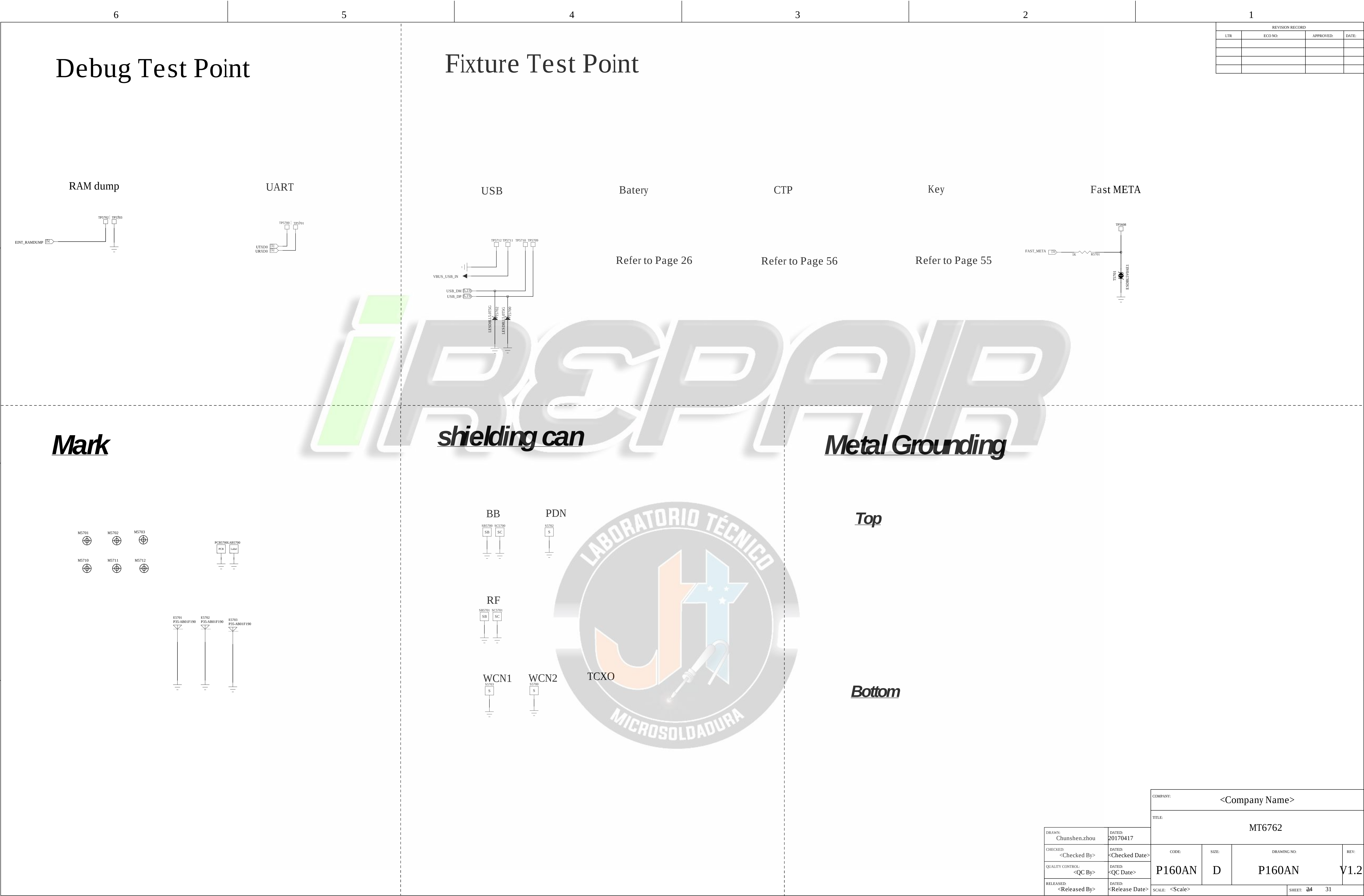


Charge Indicator

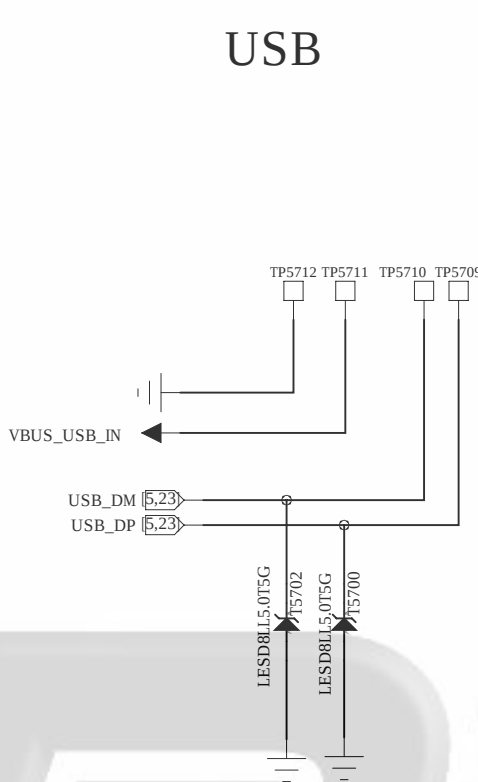
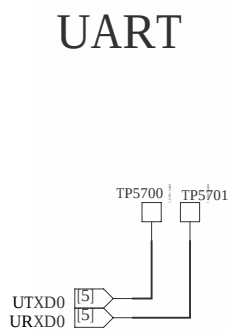
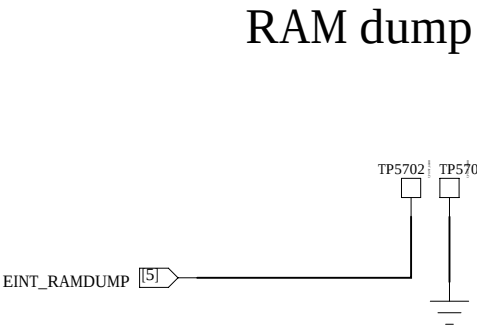


	MT6357	MT6371
R5513	NC	0 ₁
R5503	0 ₁	NC
R5504	470 ₁	NC
R5512	NC	470 ₁

		COMPANY: <Company Name>			
		TITLE: MT6762			
DRAWN: Chunshen.zhou	DATED: 20170417				
CHECKED: <Checked By>	DATED: <Checked Date>	CODE:	SIZE:	DRAWING NO:	REV:
QUALITY CONTROL: <QC By>	DATED: <QC Date>	P160AN	D	P160AN	V1.2
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>			SHEET: 32 31



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Batery

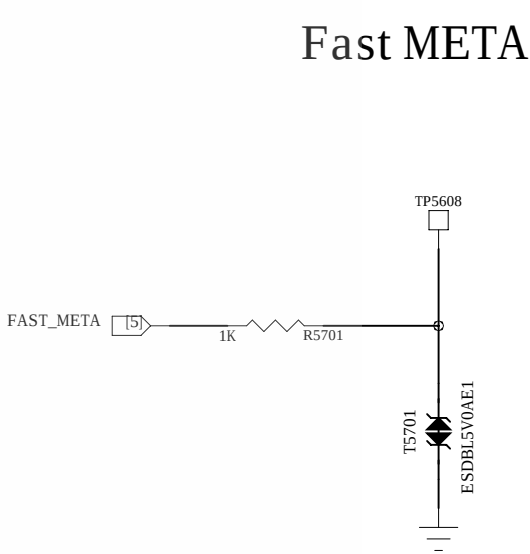
Refer to Page 26

CTP

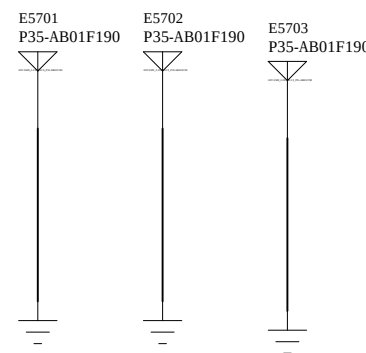
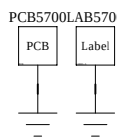
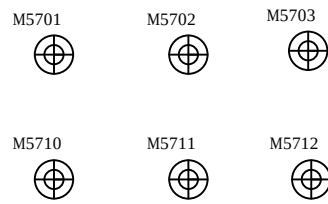
Refer to Page 56

Key

Refer to Page 55



Mark

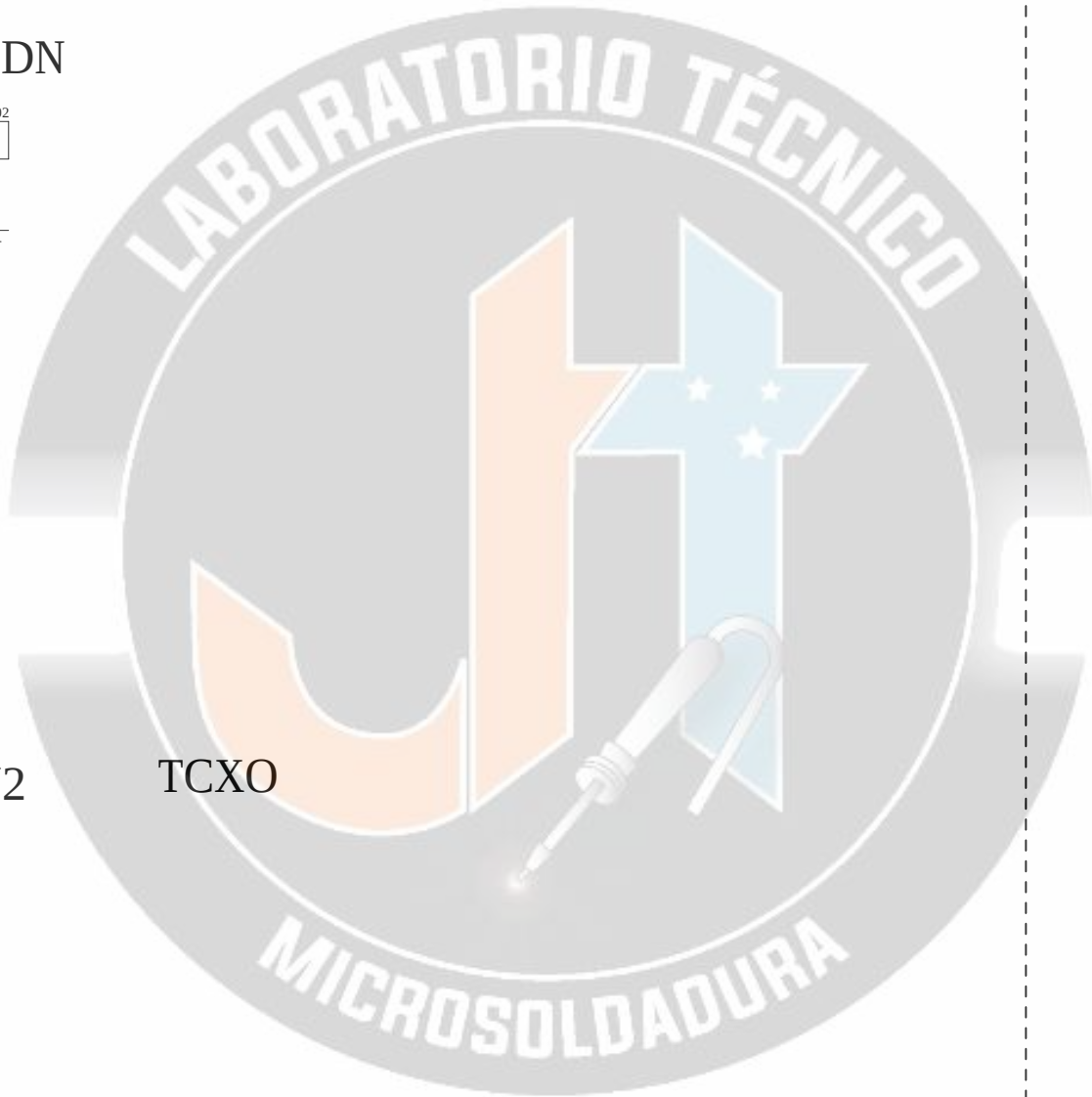
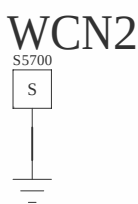
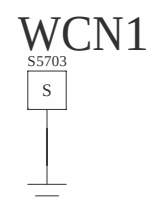
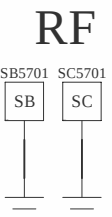
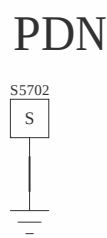
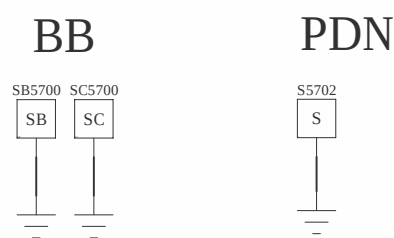


shielding can

Metal Grounding

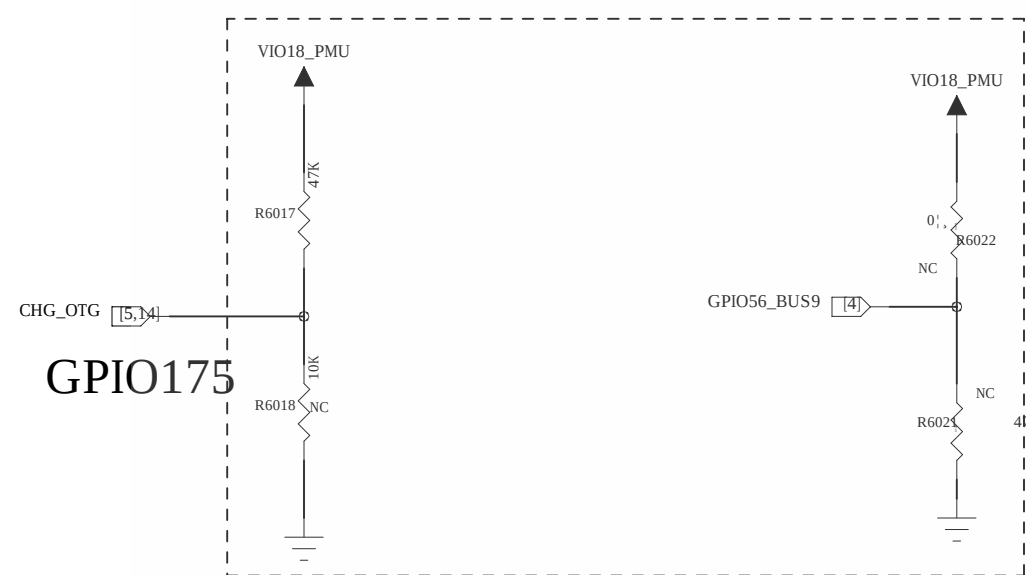
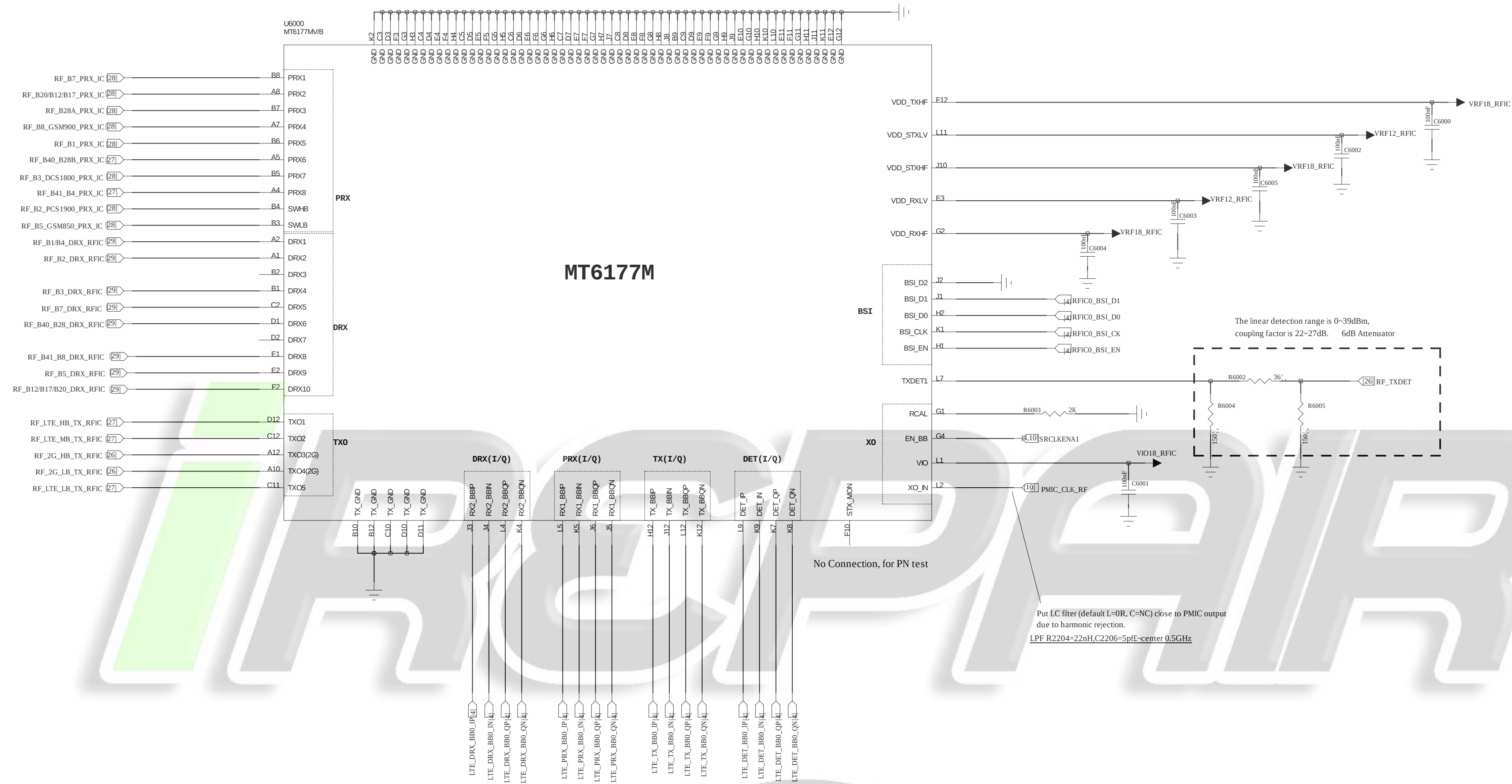
Top

Bottom

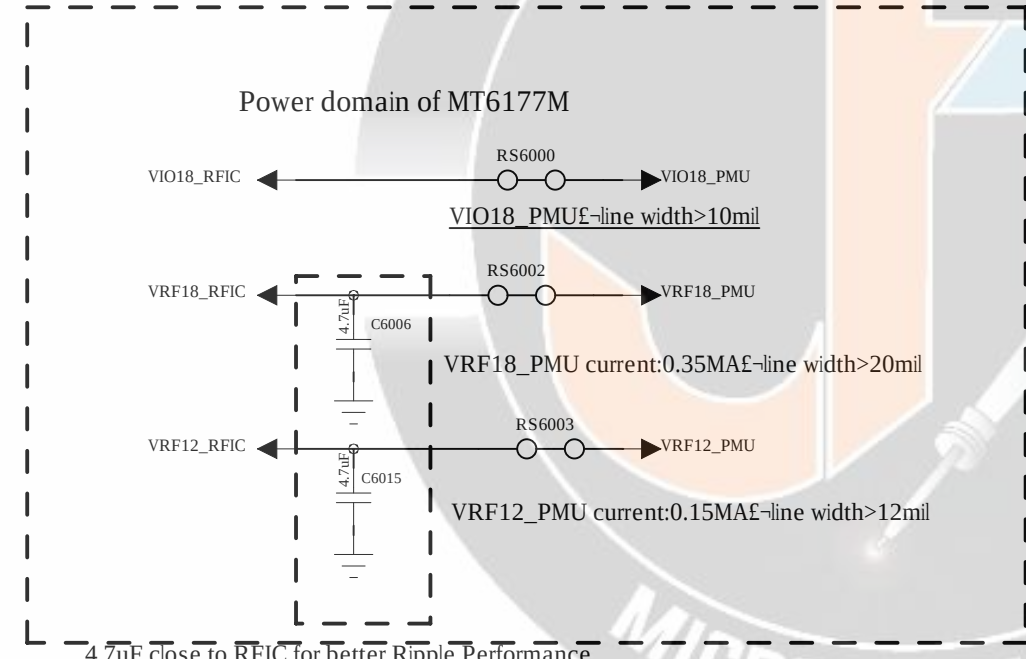


COMPANY: <Company Name>			
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CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: P160AN	REV: V1.2
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REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



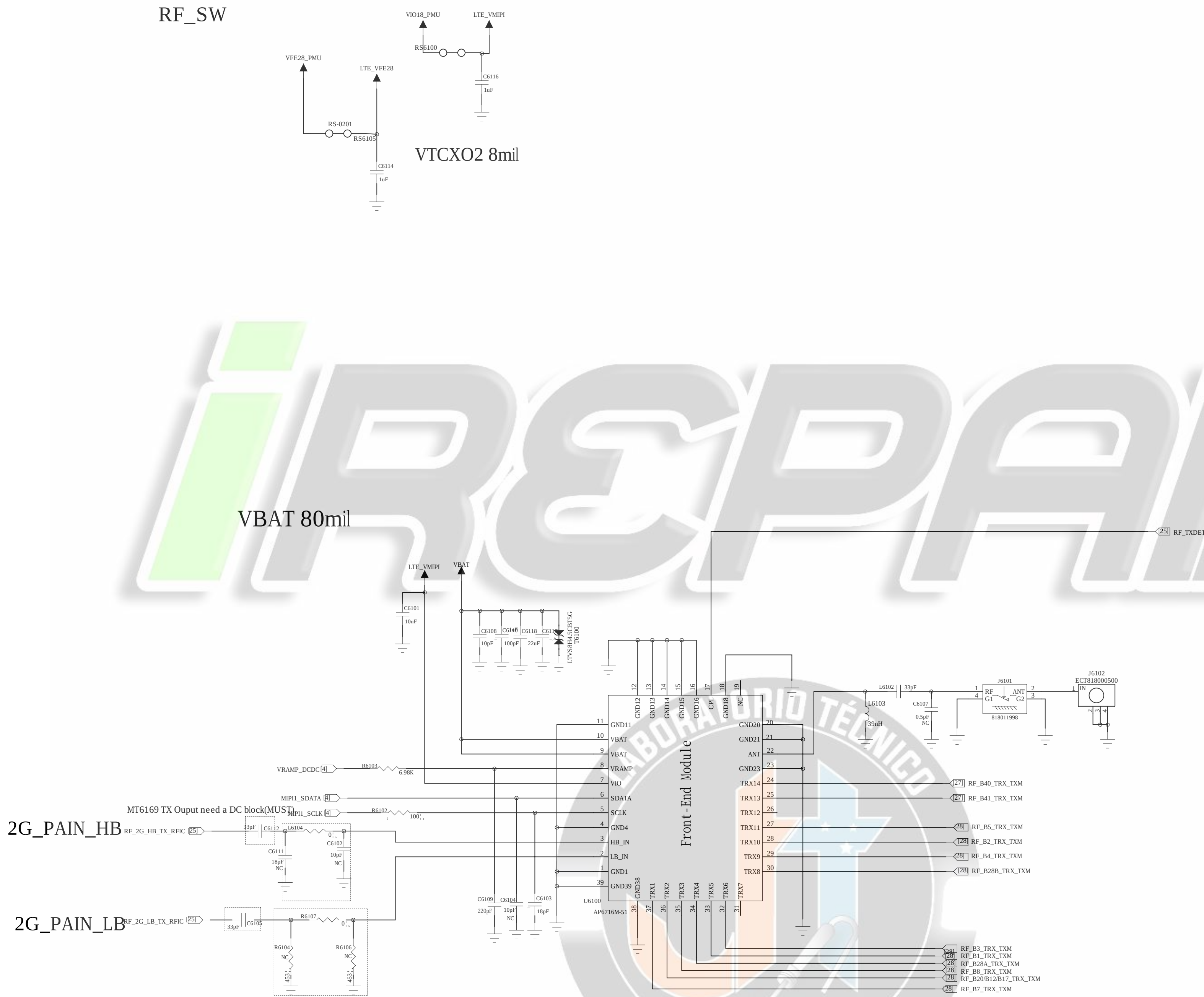
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DRAWN: Chunshen.zhou	DATED: 20170417
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REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

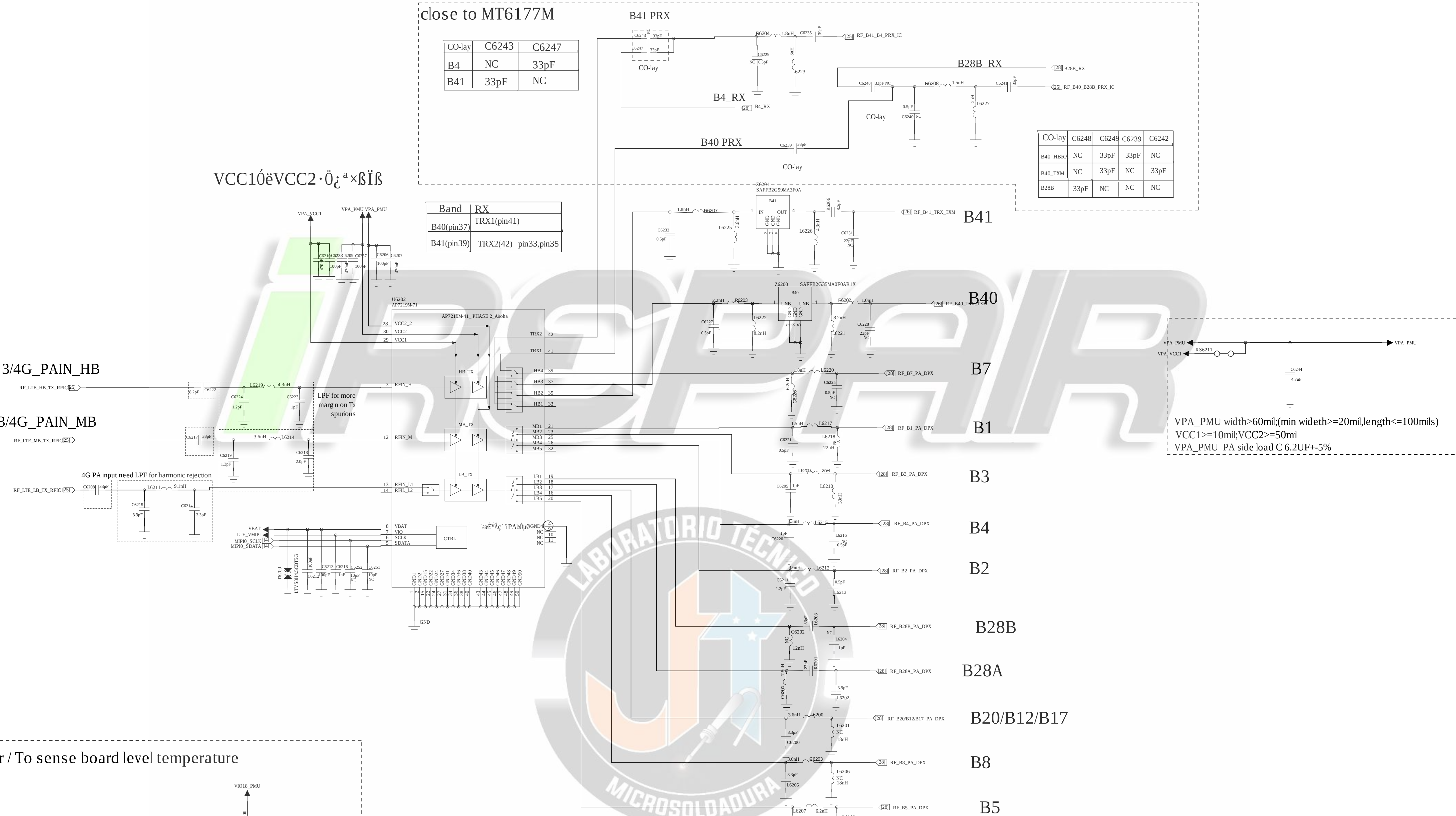


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Power Net Connection

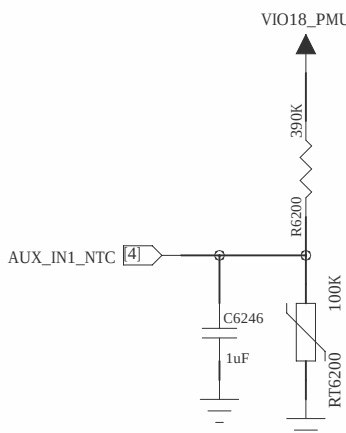
FDD LTE: B1/3/5/7/20
TDD LTE: 40
WCDMA: B1/2/5/8
GSM: B2/3/5/8
TDS: NA

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



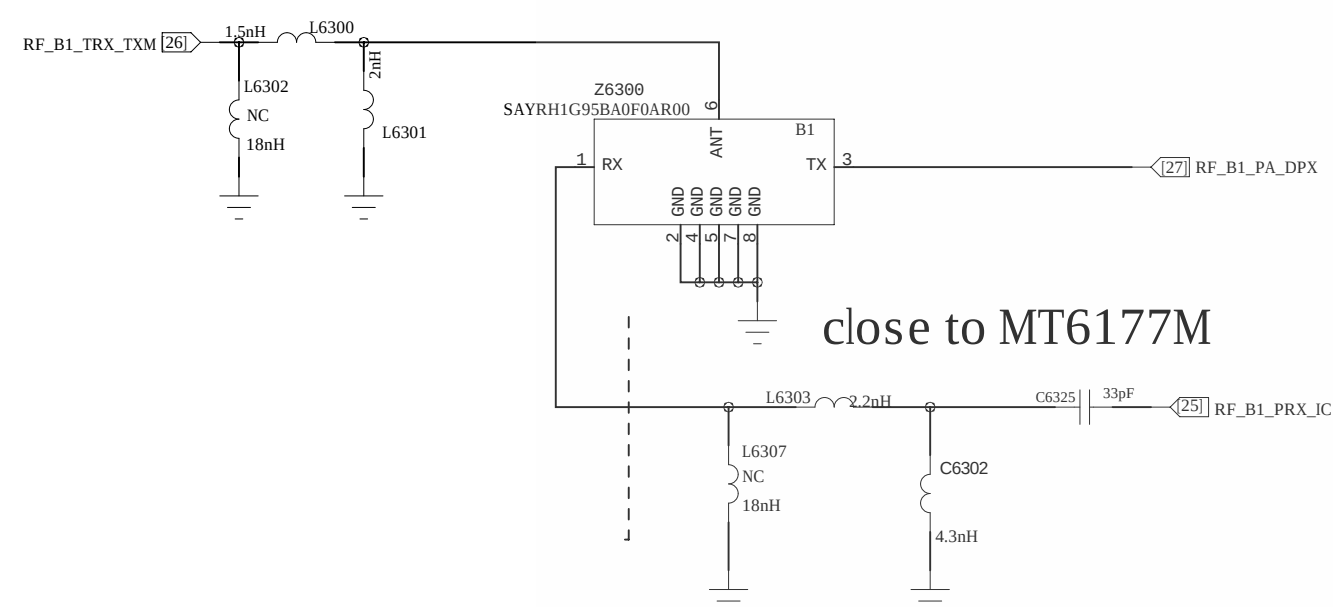
Thermistor / To sense board level temperature

Thermister have to be placed as close to B7&B1 PA as possible, routing with ground guard.
RT900 close to PA , and located in the same layer

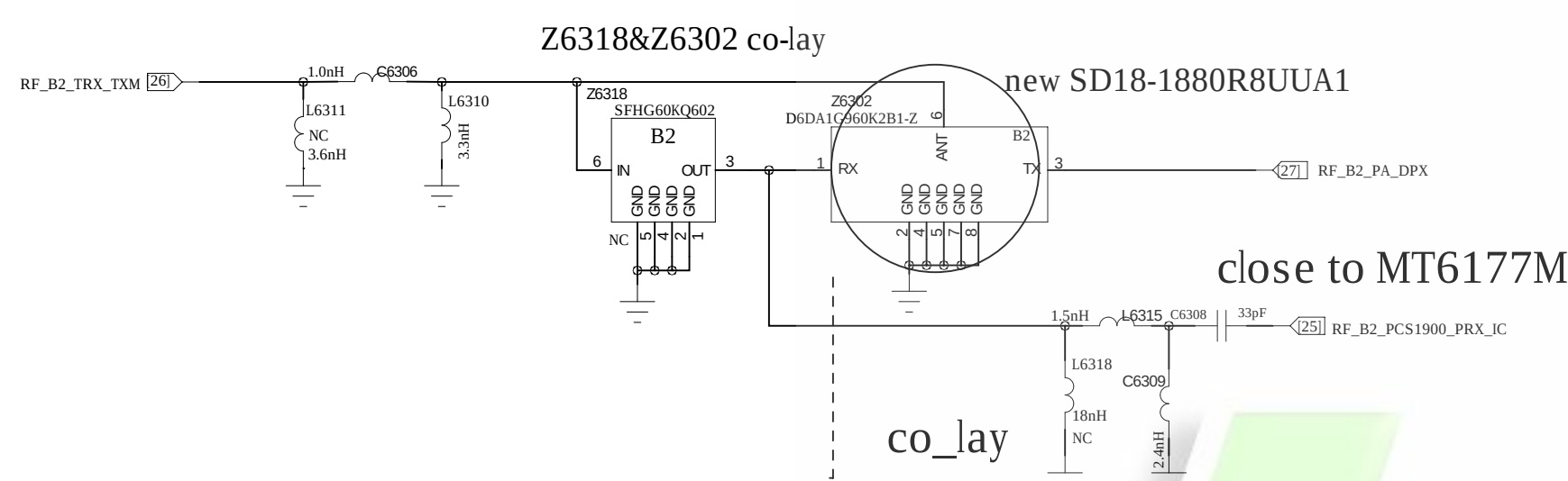


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		COMPANY: TITLE:			
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CHECKED: <Checked By>	DATED: <Checked Date>		CODE: P160AN	SIZE: D	DRAWING NO: P160AN
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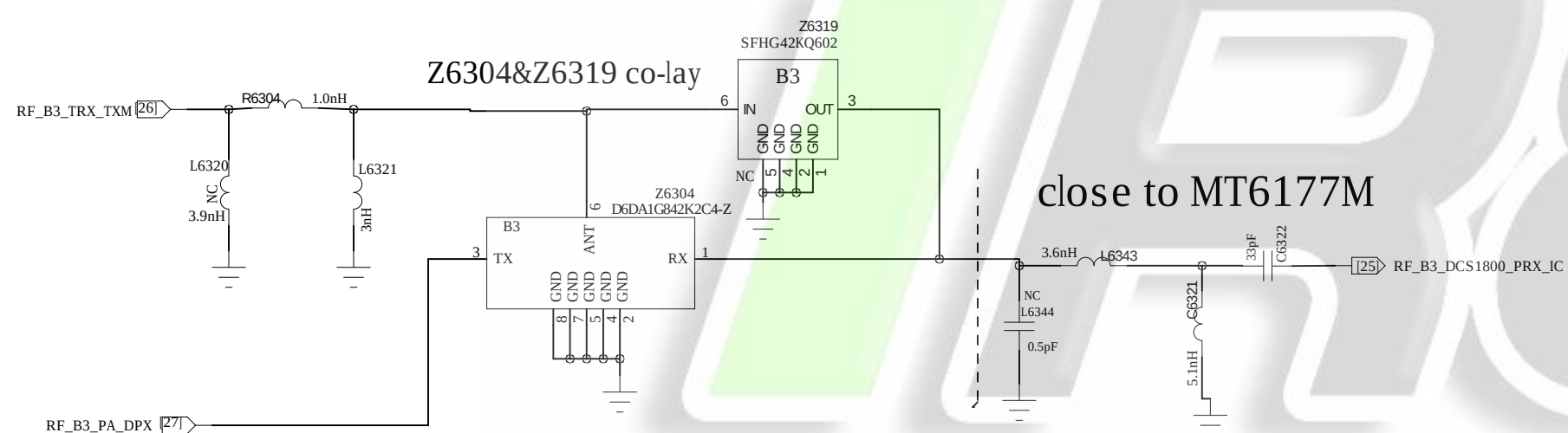
B1 TRX



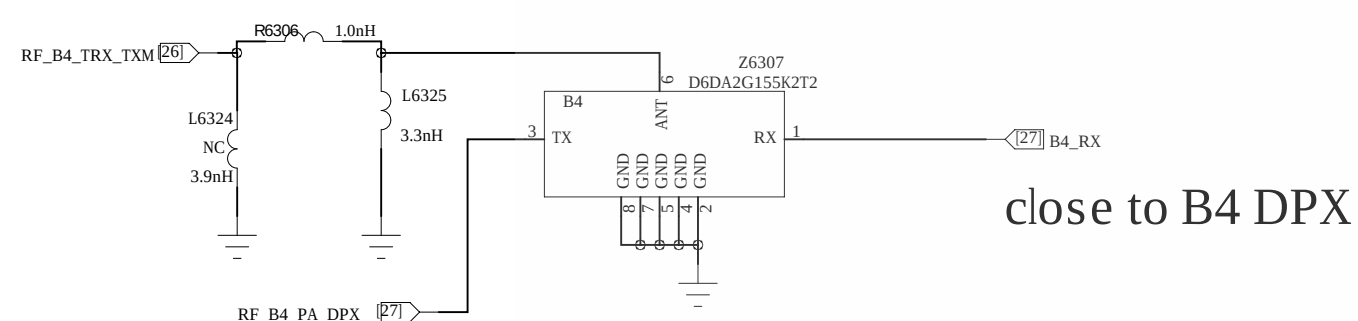
B2 TRX



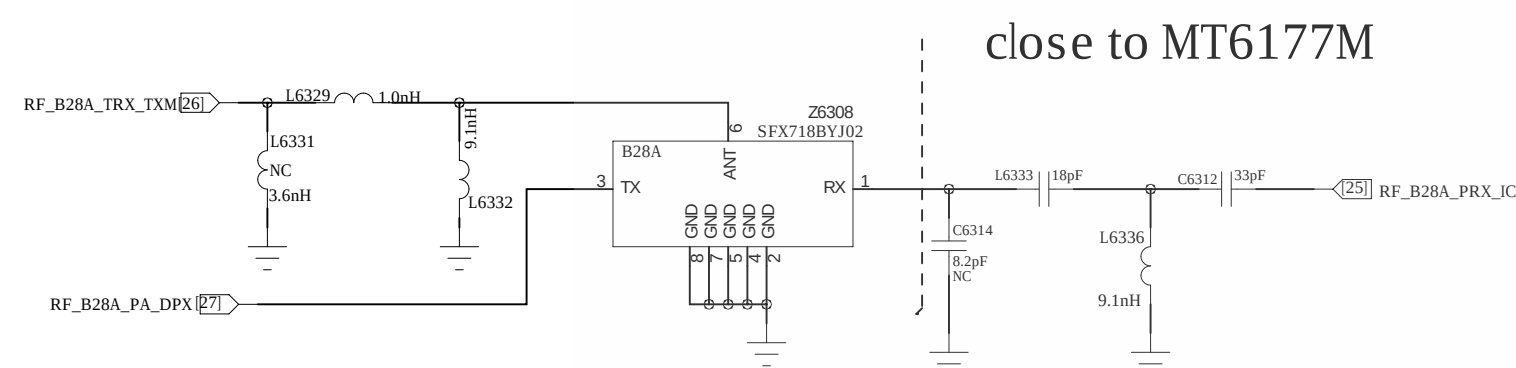
B3 TRX



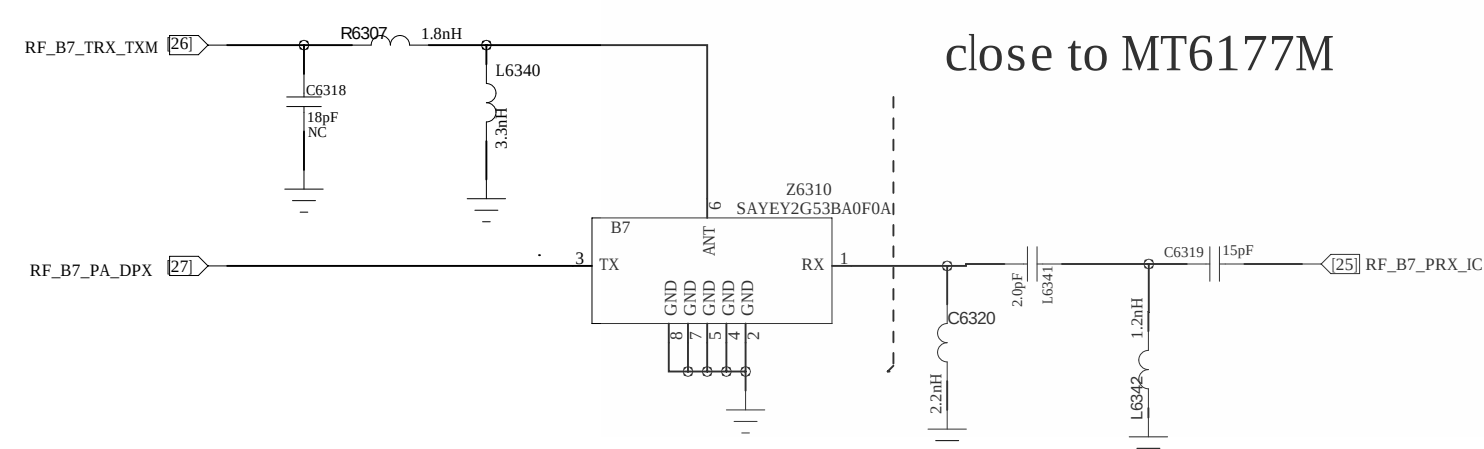
B4 TRX



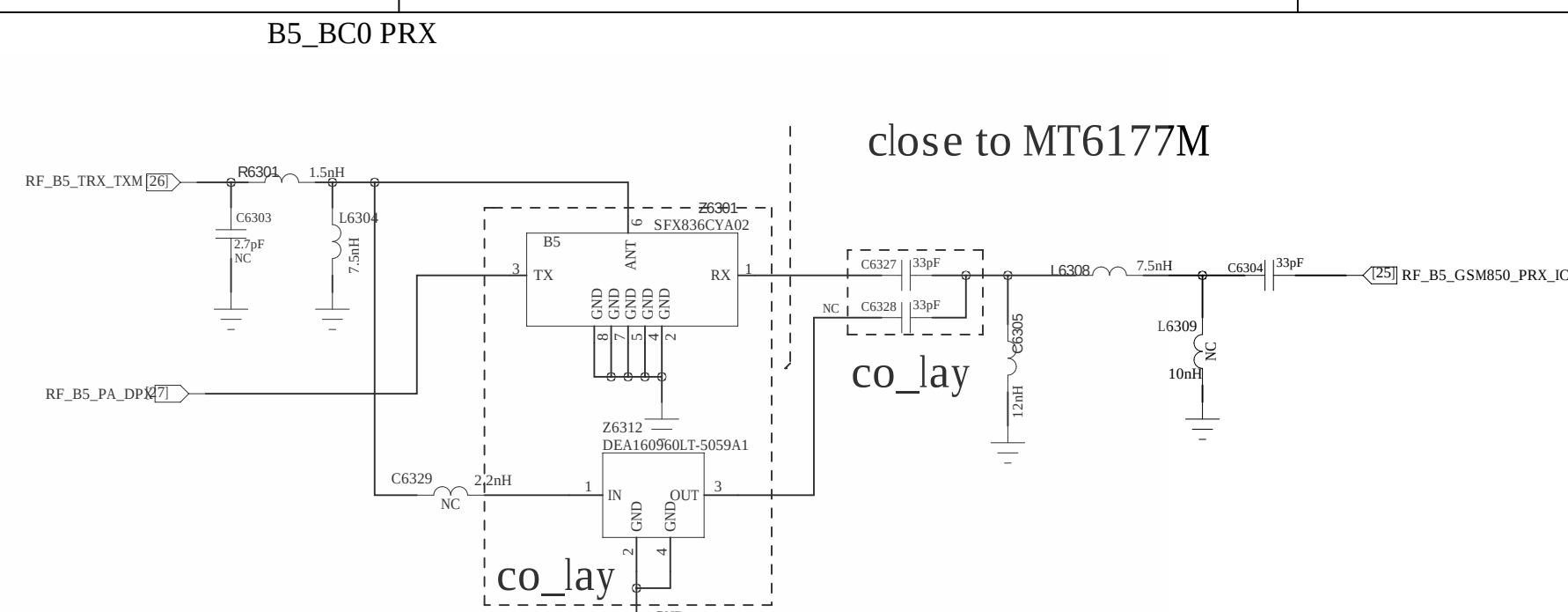
B28A TRX



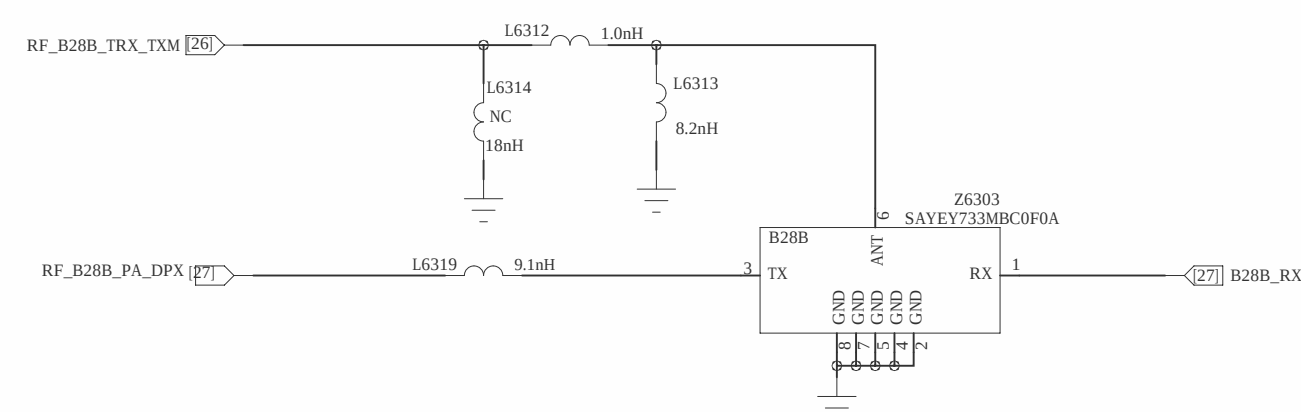
B7 TRX



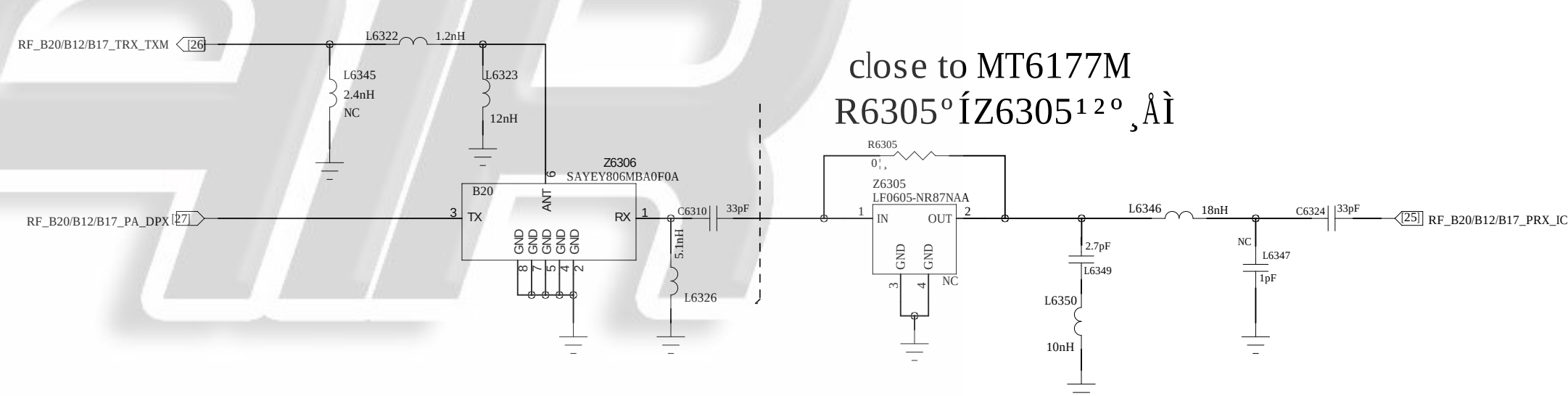
B5 TRX



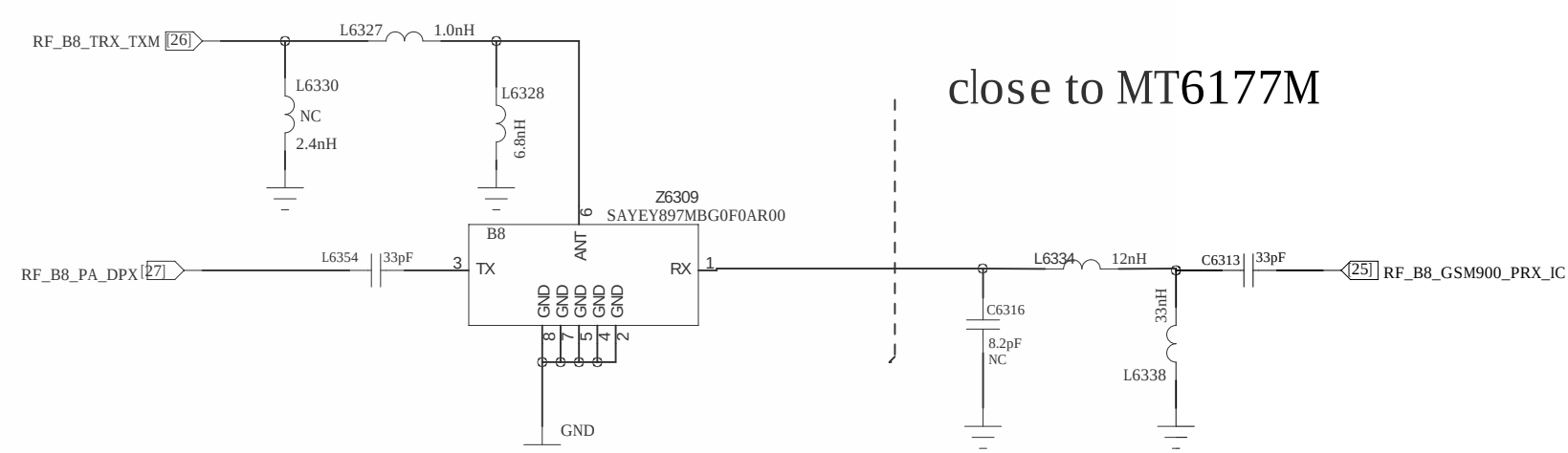
B28B TRX



B20/B12/B17 TRX



B8 TRX



GSM850 SAW

new DEA160960LT-5059A1

LBSW£°BFDEA160960LT-5059A1;C6326_33PF;C6340_NC
HBSW_PCS1900 FI168B1888GL-T;C6340_33PF;C6326_NC

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

DRAWN: Chunshen.zhou	DATED: 20170417
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY:		<Company Name>	
TITLE:			
PRI_TRX_DUPLEXER			
CODE:	SIZE:	DRAWING NO:	REV:
P160AN	D	P160AN	V1.2
SCALE: <Scale>		SHEET: 08	31

SWITCH

RFASWDH2418ATF09	RF port	V1(BUS9)	V2(BUS8)	V3(BUS7)
Band3	RF4	0	1	1
Band40/28	RF1	0	0	0
Band1/4	RF8	1	1	1
Band2	RF6	1	0	1
Band20/12/17	RF7	1	1	0
Band7	RF2	0	0	1
Band5	RF5	1	0	0
Band41/8	RF3	0	1	0

B1:DRX

close to MT6177M

B2 DRX

close to MT6177M

B3 DRX

close to MT6177M

B40 B28 DRX

close to MT6177M

B5 DRX

close to MT6177M

B7 DRX

close to MT6177M

B20/B12 /17DRX

close to MT6177M

B41/B8 DRX

close to MT6177M

COMPANY:
<Company Name>

TITLE:
DRX_ANT

DRAWN:
Chunshen.zhou

CHECKED:
<Checked By>

QUALITY CONTROL:
<QC By>

RELEASED:
<Released By>

DATED:
20170417

DATED:
<Checked Date>

DATED:
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DATED:
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CODE:
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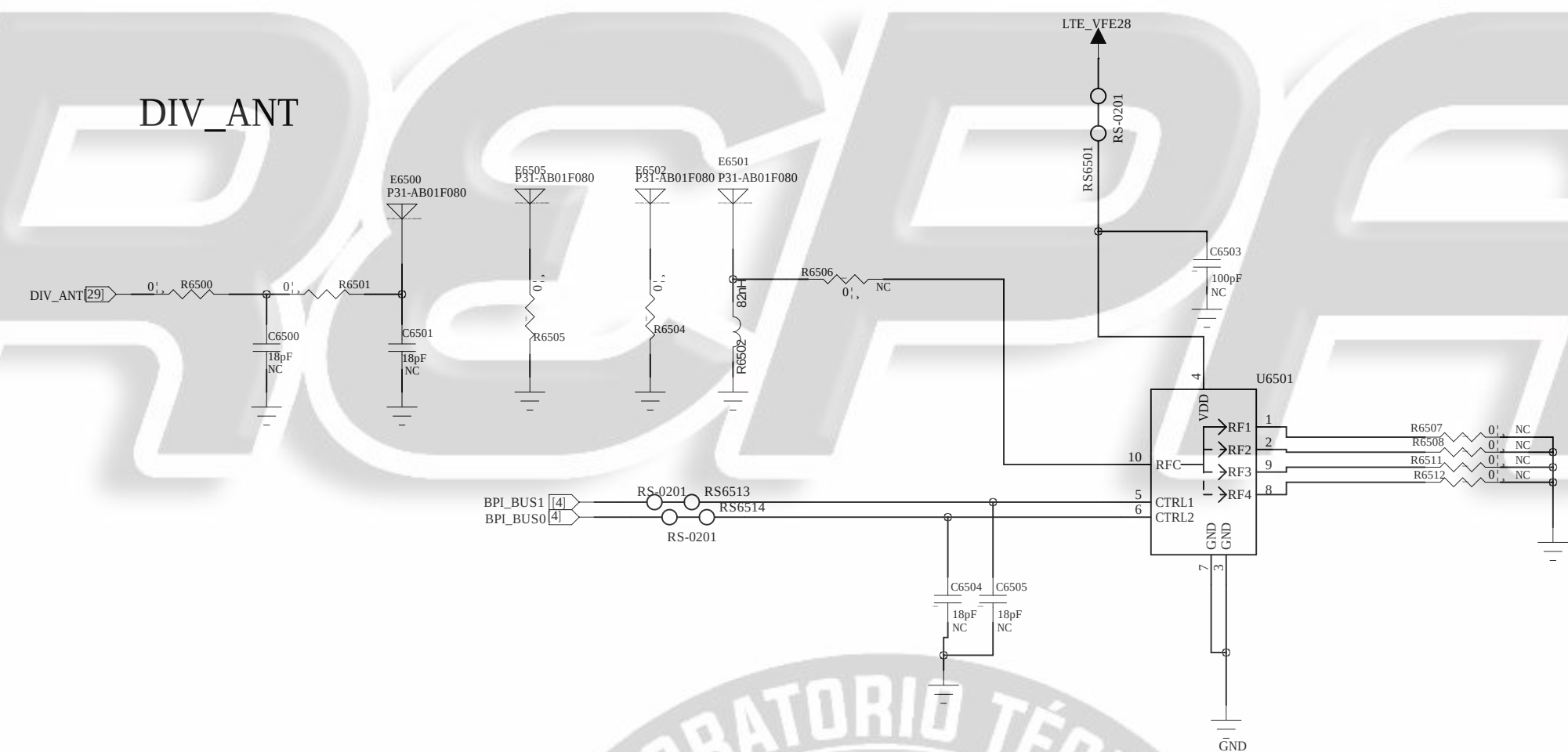
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D

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REV:
V1.2

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COMPANY:		<Company Name>	
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P160AN	D	P160AN	V1.2
SCALE: <Scale>		SHEET: 3d	31

